To Beef, or Not To Beef: Trade, Meat, and the Environment

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Abstract

More than half of European consumers report adjusting their diets for environmental reasons, yet the emissions consequences of the policies they favor remain largely unknown. I estimate a nested logit demand system for protein-rich foods from barcode-level scanner data provided by a European retailer, linking weekly purchases to product-level emissions intensities. Price is instrumented with wholesale costs and within-group shares with an advertising proxy; a fixed-margin supply side closes the model, so that cost-based policies such as tariff changes and Pigouvian taxes pass through to prices while choice-set policies such as buying local and vegetarianism operate through substitution alone. Applying the estimated elasticities to a range of counterfactuals, I show that accounting for substitution can reverse the sign of a policy's emissions effect. Buying local—a measure consumers often associate with sustainability—yields at best a small reduction and, under naive accounting, an increase, because some imported varieties are cleaner than their domestic counterparts. Removing beef and cheese instead lowers emissions by about 35%, vegetarianism by about 20%, and Pigouvian carbon taxes by up to 6%. Because I estimate rather than calibrate substitution, I also recover an across- to within-group substitution ratio well below the range assumed in quantitative trade-and-climate models, implying stronger resistance to cross-group substitution than those calibrations impose.

1 Introduction

A first order priority in the fight against global warming is to drastically reduce greenhouse gas emissions. As the agricultural sector is the second largest contributor to emissions (IPCC, 2014), food consumption behavior lies at the heart of strategies to reduce climate change. How much can demand-side policies reduce emissions from food consumption? This question is important for two reasons. First, emissions from food consumption threaten the achievement of the targets set in the Paris Agreement, even if fossil fuel emissions from other sectors had reached net zero from 2020 onwards (Clark et al., 2020). Second, more than half of consumers express concern about the environmental impact of their food choices as evidenced by a representative survey in eleven EU countries. Respondents identified measures such as buying local and reducing meat consumption as ways to decrease the carbon footprint of their diets (BEUC, 2020).

The above research question remains largely unanswered due to limitations in the granularity and scope of available data. Without detailed information on consumer substitution patterns and product-level emissions,

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it has been difficult to provide robust evidence for effective policy design. This paper addresses these data limitations by leveraging a dataset that links supermarket purchases to emissions by combining detailed retail data including product origin, transport mode, and wholesale costs with emissions data on food production and transportation. Because meat is the largest contributor to agricultural emissions (Gerber et al., 2013), the analysis focuses on weekly purchases of meat and other protein-rich products obtained through an exclusive cooperation with a European retailer. The unique product-level information allows for credible identification of demand responses and estimation of own- and cross-price elasticities using an instrumental variable strategy. Combining the estimated elasticities with emissions intensities (Poore and Nemecek, 2018), I evaluate the environmental impacts of alternative food consumption scenarios while keeping margins fixed on the retailer's supply side. I implement these counterfactuals in a partial equilibrium model of meat demand and supply.[1] This paper provides a micro-estimated demand system for food in an open economy, uses it to show that popular eco-policies like buy local can backfire while dietary changes such as cutting beef and cheese deliver reductions of up to 35%, and produces substitution elasticities that discipline calibration choices in quantitative trade and climate models.

Three stylized facts emerge from the emissions data. First, transport emissions intensities are a low share of overall emissions intensity of a product. The only case where transport emissions are relevant is with air transport. Even then, the production emissions make up the larger part of emissions. Second, production emissions intensities do not only vary across food types, but also across countries. For each food type, it would be possible to halve the emissions intensity of the most emitting country by producing with the production technology of the country with the lowest emissions intensity. Third, the variation in production emissions intensities across food types is large. On average, meat products have six times higher production emissions intensities than vegetarian products. More specifically, beef is the most polluting meat type and has a ten times higher emissions intensity than chicken, the least polluting meat type. These stylized facts show that the key question when talking about decreasing emissions from food consumption is "To Beef or Not To Beef?"

To analyze the impact of food consumption behavior on meat demand, I estimate a discrete choice model (Berry, 1994; McFadden, 1974) with the goal of calculating counterfactual scenarios to quantify the consumers' reaction to different policies. The purchasing data for estimating the model stem from a unique cooperation with a European retailer. As the retailer wishes to stay anonymous, I use the convention in the trade literature and call the country Home. I obtain weekly revenues, quantities, wholesale costs, and product characteristics like the origin of the products on a bar code level. In the words of Bresnahan (1997), obtaining marginal cost data is "rare and lucky." Here, I proxy marginal costs by the wholesale cost of the product. While wholesale costs might still be endogenous in other settings, in this context, they represent a pure economic

[1] Analyzing similar questions using trade statistics or input–output tables would make it more difficult to identify credible instruments.

cost variable without a variable markup part that might be demand-driven.[2] The quality of the data allows for an identification strategy using high quality instrumental variables for estimating meat demand. I focus on protein-rich food types[3]. The model is estimated using a nested logit specification with the food types as nests. Contrary to a standard logit specification, this allows the cross-price elasticities within nests to be different than across nests[4]. I estimate the demand parameters using instrumental variables, instrumenting for price with wholesale costs and for the within-group share with an advertising proxy. On the supply side, I summarize the retailer's pricing with a fixed per-unit margin over wholesale cost. Cost-changing policies such as tariff liberalization or a Pigouvian tax therefore pass through into consumer prices, whereas policies that only alter the choice set, such as buying local or vegetarianism, leave the prices of the remaining products unchanged and operate purely through demand-side substitution.

The regression results suggest a strong preference for domestic vs. imported goods. On top of the country of origin from the fine-print on the product, some items have a clearly visible label on the front certifying domestic production. Having such a label increases a product's market share relative to the outside good by around 50%. The elasticities derived from the price coefficient imply own-price elasticities ranging between -0.05 and -1.5. This is in line with other's findings in the meat market (Anders and Mőser, 2010; Eales and Unnevehr, 1988; Reed et al., 2003; Yang et al., 2018), and the overall pattern of beef and other ruminants being more elastic is similar to previous estimates (Lanfranco and Rava, 2014; Lusk and Tonsor, 2016).

A by-product of estimating rather than calibrating substitution patterns is that I can speak to the elasticities used in quantitative trade-and-climate models. The ratio of across- to within-group substitution I estimate is close to 0, versus 0.52–0.72 in the general equilibrium literature (Correa-Dias et al., 2026; Costinot et al., 2016; Farrokhi and Pellegrina, 2023). This implies that those models understate the resistance to substitution across food groups, which matters for any dietary counterfactual.

My main finding is that contrary to popular belief, buying local does not help, and might actually increase emissions. The buy local exercise illustrates that estimating own- and cross-price elasticities is important, because they can reverse the sign of the effect. I calculate a naive, accounting-type buy local counterfactual in the spirit of the environmental science literature (Carlsson-Kanyama and González, 2009; Eshel and Martin, 2006) with ad-hoc assumptions instead of a demand model. The assumption I make is that the quantities of each food type stay the same, but all products are produced domestically. Surprisingly, greenhouse gas emissions increase, because some imported foods have higher emissions than the domestic variety. A buy local scenario in which the elasticities guide the substitution away from imported products leads to a decrease in emissions, because on average, the consumption basket becomes less polluting. This shows that we need to know how consumers substitute in order to know whether climate policies will hurt or help the environment.

[2] I have access to the components of the wholesale costs for a subset of the data and can verify this.

[3] Specifically, poultry, pork, beef, veal, lamb, fish, cheese, eggs, legumes, and tofu

[4] In intuitive terms, the cross-price elasticity between domestic and imported beef is allowed to be higher than the cross-price elasticity between beef and chicken.

As a counterpoint to the buy local (autarky) scenarios, another trade-related counterfactual considers free trade. To account for tariffs and quotas, I provide two free-trade scenarios. The first assumes tariffs on most products. The second assumes a binding quota and uses the ad-valorem equivalent tariffs to calculate the counterfactual. Both scenarios lead to a slight increase in emissions. As in many developed countries, beef, lamb and veal have the highest tariffs, thus a decrease in tariffs under a free trade scenario leads to more consumption and overproportionally more emissions. For some products, both price and quantity decrease. This happens due to a simultaneous price change in several products where the cross-price channel dominates, as detailed in the intuition section.

The two other groups of scenarios revolve around reducing meat consumption like vegetarianism or a no-beef no-cheese scenario and Pigouvian taxes of 50, 100, and 150 EUR/tCO_2e (tons of CO_2 equivalents) that are proportional to the emissions intensity of each product. These vegetarian and Pigouvian tax scenarios lead to a larger decrease in emissions than buying local. Vegetarianism results in a decrease of emissions of around 20%. A related counterfactual considers decreasing beef consumption while all other meat products can be purchased. Since beef is the most polluting product, refraining from it yields a decrease in emissions of 14%. The second largest contributor to total emissions is not a meat product, but cheese. Per kilogram, the emissions intensities of cheese are comparable to those of fish and better than pork or chicken. A scenario without consumption of beef and cheese, but allowing all other meat types leads to a decrease of around 35% in emissions and thus results in a larger decrease than vegetarianism. Thus, the answer to the question posed in this paper is "Not to beef," at least from an environmental perspective. The Pigouvian tax counterfactual with the lowest level of taxes of 50 EUR/tCO_2e leads to a decrease of around 2%, while the other two scenarios lead to around 4% and 6%, respectively. In these scenarios, all prices increase, because all products are associated with some level of emissions.[5] The prices of greener products increase less, thus there is substitution of consumption towards these greener products. Again, for some products the cross-price channel dominates and both price and quantity increase.

I provide two robustness checks: I vary the definition of the outside good and I perform the analysis on the barcode level data. I find similar results for both exercises. To assess relevance and external validity, I examine three aspects. First, by holding consumption shares constant and assuming the lowest possible emissions intensity for each food type, I find that emissions could decrease by 9% if production practices improved. This number serves as a lower bound, indicating greater emissions reduction potential from consumption-based policies. Second, I analyze the buy local scenario, considering extremes where Home has the highest and lowest production emissions intensity. If Home were the least efficient, emissions would rise by 34%; if the most efficient, they would decrease by 10%. This suggests that buying local might lead to a small decrease in emissions for greener countries, and a large increase in emissions for dirtier countries. Thus, vegetarianism and Pigouvian taxes seem like more effective policies for decreasing greenhouse gas emissions

[5] In a general equilibrium model, the tax revenues would be rebated lump-sum to the households. Since this is partial equilibrium, this mechanism is not explicitly modeled in this setup.

from food consumption. Third, differences in consumption patterns across countries are mostly due to prices and product traits, with minor variations in preferences. Using European meat consumption data, I find that Home's patterns align with those of other countries. Based on these considerations, I conclude that my results are likely to be representative for most developed European countries.

This paper lies at the intersection of international trade and environmental economics. While the theoretical environmental-trade literature has long emphasized that the demand side matters, empirical evidence beyond automobile markets is scarce (Antweiler and Gulati, 2016; Cherniwchan et al., 2017; Copeland and Taylor, 1995; Davis and Kahn, 2010; McAusland, 2008). An active area of research analyzes the effects of trade and climate in general equilibrium models, focusing on agricultural production (Conte et al., 2021; Correa-Dias et al., 2026; Costinot et al., 2016; Farrokhi et al., 2025; Farrokhi and Pellegrina, 2023; Gouel and Laborde, 2021; Hsiao, 2026). This paper provides the first micro-estimated demand system for food in an open economy. The substitution elasticities in this study can help discipline calibration choices in quantitative trade-and-climate models. Unlike Shapiro (Shapiro, 2020), who documents that greener industries face higher tariffs in the US, I find the opposite pattern in Home's agricultural sector, so free trade increases emissions in this setting.

In the environmental sciences, several papers analyze the emissions associated with food products along the supply chain. These include accounting exercises comparing the emissions of different dietary choices (Carlsson-Kanyama and González, 2009; Eshel and Martin, 2006; Scarborough et al., 2014), analyses of the relative importance of transport versus production emissions (Avetisyan et al., 2014), and supply-side studies of dietary shifts (Popp et al., 2010). My findings are broadly consistent with this literature, which generally concludes that transport accounts for only a small share of the total emissions of food products. Studies evaluating the environmental impact of buying local typically find modest reductions in emissions. Some authors emphasize that shifts away from meat consumption could reduce emissions (Poore and Nemecek, 2018; Weber and Matthews, 2008). These studies typically evaluate counterfactual scenarios that impose changes in production or consumption, e.g. reducing meat production by half. My analysis contributes to this debate by estimating the substitution patterns of consumers and applying them in alternative policy scenarios rather than imposing them exogenously.

This paper analyses the effect of food consumption behavior on the environment. I apply methods from the industrial organization literature (Berry, 1994; Nevo, 2001) with high-quality purchasing data to a question that is important both for international trade and environmental economics. Although the industrial organization literature has examined the effects of food consumption on health and other outcomes (Briggs et al., 2013; Dubois et al., 2017; Edjabou and Smed, 2013; Griffith et al., 2019), this is the first study to assess its impact on climate change within an open economy. This perspective allows me to compare buy local, which might even hurt the environment, to Pigouvian taxes, which deliver substantial reductions in emissions. In contrast to supply-side studies that evaluate how environmental policies affect upstream emissions (Dominguez-Iino, 2021), my work assesses demand-side policy implications, demonstrating that accounting for consumption behavior is essential to effective climate policy.

This paper is organized as follows. Section 2 describes the data. A set of stylized facts are presented in section 3. Section 4 outlines a model of meat demand, and section 5 discusses estimation and presents the regression results. The counterfactual scenarios are presented in Section 6. Section 7 discusses robustness and external validity and Section 8 concludes.

2 Data

Through an exclusive cooperation with a European retailer, I obtain confidential scanner data on revenues, quantities, wholesale costs, and country of origin for protein-rich food products at the bar code level. As the retailer wishes to remain anonymous, I follow the convention in the international trade literature and refer to the country as “Home.” The retailer has a very high market share for food and is the leading seller of meat products in Home. To protect the retailer’s anonymity, I also anonymize all other European countries. The robustness section addresses the concerns associated with this anonymity.

The key advantage of the scanner data is that they allow me to observe wholesale costs and the country of origin of all products. Throughout the analysis, I treat the wholesale cost as a proxy for marginal cost. For a subset of products, I observe the cost calculation and can verify that it reflects underlying production costs rather than demand-driven components. This information is crucial for identification because retail prices are endogenous: they respond to demand shocks. The observed wholesale costs therefore provide product-level cost shifters that can be used as instruments for prices when estimating consumer demand. Such instruments are typically unavailable in more aggregated data sources such as trade statistics or input–output tables. Moreover, because the data capture purchases by retail consumers, the estimated elasticities reflect household demand rather than processor demand, which would also include agricultural inputs used by restaurants and food processors. The information on the country of origin further allows me to distinguish between domestic and imported products when estimating substitution patterns relevant for trade policy.

The analysis focuses on protein-rich foods because they account for a disproportionate share of food-related greenhouse gas emissions and constitute a natural substitution group in consumer diets. Substitution between different protein sources such as beef, poultry, fish, and plant-based proteins is therefore a key margin through which dietary choices affect emissions. In addition, most protein products in the data are relatively close to commodities, which implies that wholesale costs closely reflect underlying production costs and can serve as credible instruments for retail prices. Focusing on proteins therefore sharpens the environmental relevance of the analysis while improving the identification of consumer demand.

There are around 12000 bar code level products from 37 countries in the period of 2017-2019 with a weekly frequency. Product characteristics include country of origin, labels (e.g. organic), brand name (e.g. “European Retailer Premium Brand”), food type (e.g. beef or cheese), sales or promotions, and product category (e.g. ground meat). The data consists of all products that contain exactly one of the food types considered (beef, veal, lamb, pork, poultry, fish, crustaceans, cheese, eggs, tofu, legumes). I exclude products that are a combination of two food types and calculate the weekly average price as revenue divided by quantity.[6] Prices and quantities vary by product, week, and region. The retailer operates in all of the different geographical regions of Home.

I consider both emissions from production and from transportation in the analysis. Production emissions intensities stem from a peer-reviewed meta data of life cycle assessments of food products (Poore and Nemecek, 2018). The subset of the data I am using takes into account emissions related to the majority of the value chain of food production. It covers land use change, crop production for animal feed, raising livestock, processing, packaging, retail, and losses. These production emissions intensities are expressed in kgCO_2 equivalents per kg of edible product and vary by food type (beef, beef dairy herd, lamb/mutton, poultry, pork, crustaceans, fish, tofu, eggs, cheese, legumes), and country. Appendix section 10.1 gives more details on how I combine the data sets.

The transport emissions intensities originate from two different sources. Emissions intensities from air transport come directly from the European retailer. An internal study calculates the greenhouse gas emissions of the transport by airplane in kgCO_2e per kg of each product that is flown. This data takes into account distances, weight, and whether the product was transported as belly freight or not. I calculate emissions intensities from other modes of transport (truck, rail, ship) using shares for each mode. For some products, I obtain the mode of transport from the European retailer. For the other products, I work with average modes of transport per food type obtained from customs data. Emissions intensities for each transport mode come from Poore and Nemecek (2018) and are given in kgCO2e per gram of product transported for 1km. I calculate the distances using great circle distances from the Center for International Prospective Studies (Mayer and Zignago, 2006).

For estimating the model, I need to define an outside good. This outside good captures the option not to buy any of the protein-rich products in my data. I define the outside good to be any other food type. This definition allows consumers to substitute away from meat to any other food product - be it a meat substitute or some potatoes. Thereby, I make use of the empirical pattern that most humans eat around 1.8 kg of food every day to define the market size. I obtain data from a representative food consumption survey in Home to match the quantity of the outside good to the quantity consumed by the average Home resident. This results in setting the market share of the outside good to around 85%. I assume that the outside good is domestic and thus, the transport emissions are zero. As for the production emissions intensities, I use the average of all other food types (without drinks) in Poore and Nemecek (2018) that are not available in the data set from the European retailer on prices and quantities. This includes the emissions from fruit, vegetables, and starches.

[6] Verifying this calculation with a subset of the data for which exact prices are available shows an accuracy of over 99%.

To match the level of variation of the emissions data and to simplify estimation, I aggregate the bar code level products by food type, continent of origin, and transport mode (flight, truck, rail, ship)[7]. However, the underlying product-level data remain essential for estimation. They provide the price and cost variation used to identify demand elasticities and allow the construction of category-level prices that reflect substitution patterns across products with different origins. I treat Home as its own continent and aggregate the other countries into continents according to seven regions as defined in the World Bank Development Indicators. Leaving the country of origin on the country level instead of aggregating to the continent level would lead to a lot of missing values, as not every food type is represented for every country in the data set on production emissions intensities. This results in a data set with 38 distinct products varying across regions and weeks. Whenever I refer to a "product," I refer to this aggregated product definition unless otherwise stated. I report regressions for both the aggregate and the barcode level product definition.

Table 1 shows a decomposition of the total price variation, which attributes the explained variance fairly across correlated predictors by averaging their incremental contributions over all possible orderings in a regression model of all shown predictors on the price. The results show that wholesale cost and product fixed effects together explain the vast majority of price variation—about 90% combined—while temporal and regional factors contribute only marginally. This highlights that price differences are primarily driven by product-specific characteristics and underlying cost structures rather than time or location effects. Around 10% of the price variation comes from sales. There are three types of sales. With standard sales (sales 1), the price is reduced per kg of product. Packaged sales (sales 3) also offer price reductions, but at the same time require the consumer to buy the product in packages (e.g. three sausages instead of one). The third type of sales only occurs when a product is close to expiry. In that case, a product might be discounted with stickers (sales 2).

Table 1: Decomposition of Price Variance by Explanatory Variable

Variable	% of Price Variance Explained
Wholesale Cost	47.1
Product ID	45.0
Sales 3	3.5
Week-Year	1.5
Sales 2	1.5
Region	1.1
Sales 1	0.4

Note:
The table shows the share of total price variation explained by each variable based on Lindeman et al. (1980).

Table 2 gives some product characteristics by food type[8]. Overall, most food is not imported, thus imported

[7] I also provide estimates on the barcode level in the Appendix.

[8] To respect the retailer's anonymity, I am limited in the types of summary statistics I show in Table 2.

market shares are low.

Table 2: Summary Statistics

Food Type	Imported	Continents	Quantity	Sales 1	Sales 2	Sales 3	Label Organic	Label Local
beef	no	1	2164	15	9	4	12	92
beef	yes	5	222	4	38	7	<1	0
cheese	no	1	8549	5	9	<1	4	64
cheese	yes	1	689	3	3	<1	1	0
eggs	no	1	2722	<1	5	<1	19	97
eggs	yes	1	569	<1	2	<1	<1	0
fish	no	1	714	7	18	4	11	6
fish	yes	6	360	4	27	7	15	0
lamb	no	1	204	6	12	5	<1	50
lamb	yes	2	161	8	35	10	2	0
legumes	no	1	517	<1	10	<1	12	<1
legumes	yes	1	143	<1	<1	<1	3	0
pork	no	1	4141	9	14	2	2	85
pork	yes	1	7	2	<1	6	0	0
poultry	no	1	4055	6	19	3	1	73
poultry	yes	1	90	4	23	9	0	0
tofu	no	1	246	<1	18	2	20	5
tofu	yes	1	64	<1	0	3	41	0
veal	no	1	80	4	12	2	<1	80
veal	yes	1	<1	4	0	<1	0	0

Note: This table shows the summary statistics of the aggregated products. Each row is a food type split by import status, aggregating all barcodes of that food type and origin. 'Imported' indicates whether the products are imported, and 'Continents' is the number of continents the products are sourced from. Each product is imported from at least one continent. Continents are defined according to the World Bank Development Indicators. The quantity is expressed in tons per 1 mn consumers. Sales are in % of revenue. Sales 1 refers to standard sales where the price is reduced per kg of product. Sales 2 only occurs when a product is close to expiry. Sales 3 refers to packaged sales which require the consumer to buy the product in larger packages. Labels are in % of quantity. <1 means between 0 and 1, but >0.

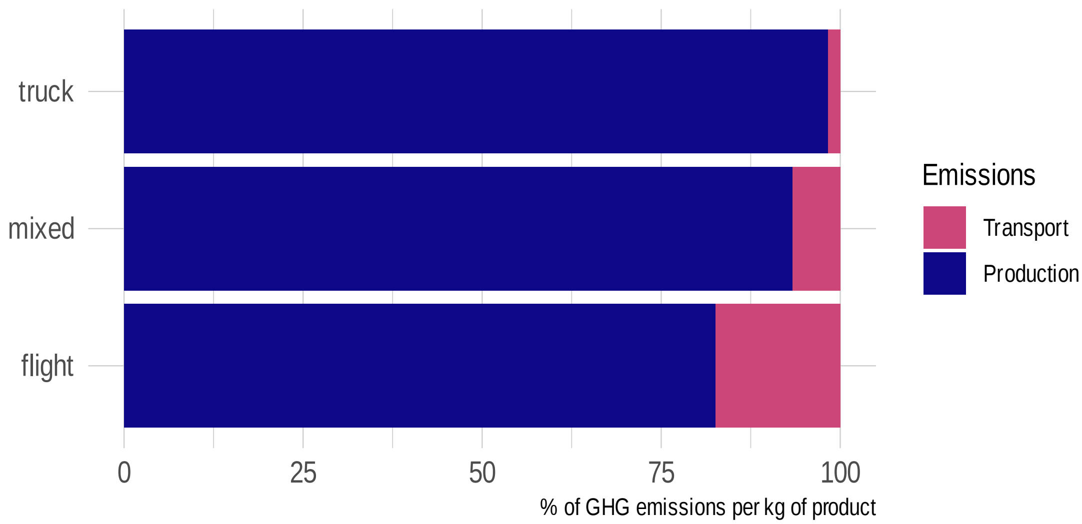
For the trade regime analysis, I rely on the experts at the European retailer and with their help, categorize the barcode level products according to which trade regime they belong to. There are three types of trade regimes: no restriction, tariff, and tariff-rate quotas. I collect data on the trade restrictions for different products from the trade statistics of Home. Additionally, I gather the ad valorem equivalent (AVE) applied tariff from the Market Access Map (ITC, 2020) of the World Trade Organization. The AVE exceeds the binding within-quota tariff, so the scenario based on it provides an upper bound on the free-trade price change, as discussed in Section 6.3.2.

3 Stylized facts

One wide spread belief is that regional foods are more sustainable than imported ones. This "buy local" concept relies on two implicit assumptions. The first is that transport emissions are important for the overall emissions of a product. The second is that emissions from production are the same across countries.

Combining consumption data from Home with production emissions intensities (Poore and Nemecek, 2018) and transport emissions intensities, Figure 1 shows that transport emissions intensities only account for a

Figure 1: GHG emissions intensities of selected imported meat products



Note: Each horizontal bar corresponds to one transport mode (truck, mixed, air) and shows the product with the median total emissions intensity in that mode: the truck product is pork, the mixed and air products are beef. Each bar is split into the share of the product's per-kilogram GHG emissions coming from production and from transport to the retailer. The x-axis is the % of that product's total per-kg emissions.

small fraction of overall emissions intensities of products consumed[9]. The three products represent median overall emissions intensity per transport mode. A mixed transport mode might be transported via truck, rail, or ship. The product coming only by truck is pork, the other two are beef. While choosing products with higher or lower production emissions intensities would change the proportions of transport and production emissions, the message would be the same: Transport emissions do not play an important role in overall emissions. Appendix Figure 13 illustrates this for the case of beef and tofu. Therefore, the first assumption of the buy local concept is flawed.

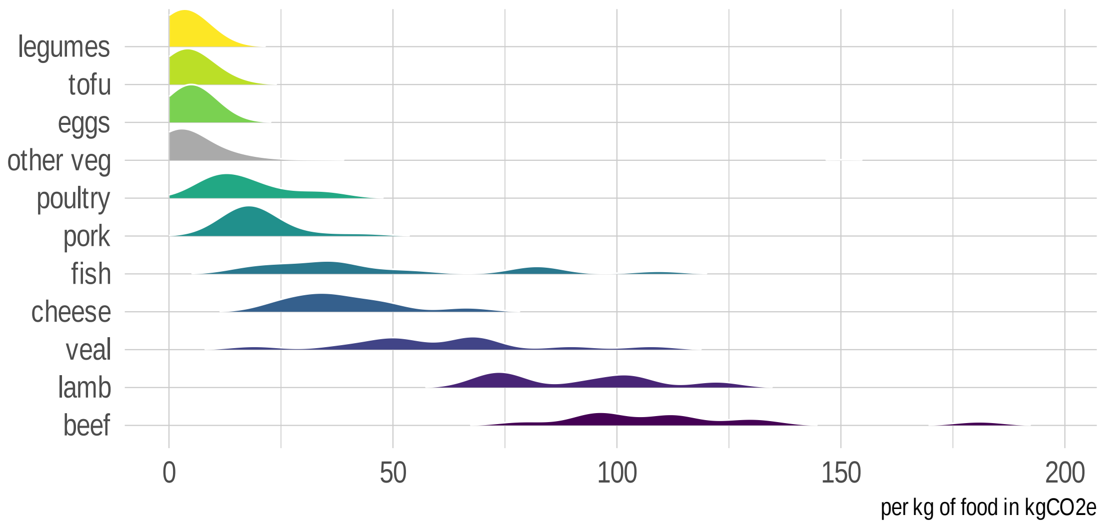
Figure 2 shows the distribution of emissions of producing 1 kg of food across different countries[10][11]. There are three main takeaways from this graph. First, the emissions intensities of vegetarian products are lower than those of meat products. Therefore, shifting away from meat products can lower emissions. Second, the production emissions intensities vary substantially even within meat products. Substituting beef by poultry or pork could reduce emissions intensities by a factor of 10. The reason for these stark differences is a biological one: Ruminants (cattle, lamb, mutton) have higher emissions intensities because of lower efficiency when

[9]Transport emissions refers to the transport of the final product to the retailer. Transport emissions during the production process are very small and not accounted for in this analysis.

[10]These emissions intensities include several stages of the life cycle of a product, including land use change. Depending on whether one views deforestation as a fixed or a sunk cost, land use change should not enter in the environmental cost. As shown Appendix Figure 14, excluding land use change results in lower overall emissions intensities for all products, but the ordering and the spread of emissions intensities across meat types stays the same.

[11]The production emissions data is very similar when considering emissions per calories or emissions per grams of protein.

Figure 2: GHG emissions intensities across countries



Note: Each ridge is a food type. Its shape is the distribution of production GHG emissions intensity across producing countries, so the horizontal spread captures cross-country variation within a food type. Food types are ordered by mean intensity. The x-axis is emissions intensity in kgCO_2e per kg of food.

converting feed to edible meat (feed-conversion ratio). Third, the emissions intensities vary across countries. For each food type, production emissions intensities could be at least halved if production took place with the technology of low-emissions intensity countries. This suggests that the second assumption of the buy local rule does not hold either.

The figures show that what we eat is much more important than whether our food is local. All this evidence suggests that the fundamental question when talking about decreasing emissions from food consumption is "To Beef or Not To Beef."

4 A discrete choice model of meat demand

The previous sections described the data and presented some stylized facts. In this section, I outline a discrete choice model of meat demand. Estimating demand is the basis for gauging the effect of different policies. The goal of this exercise is to estimate consumer's reaction to changes in choice sets and prices which enables me to conduct counterfactual exercises. The model helps to estimate own- and cross-price elasticities of meat products using a nested logit framework (Berry, 1994). The nested logit structure allows for flexible substitution patterns within meat categories while remaining tractable with market share data.[12] First, I present the theoretical model on the demand side. Then, I add the supply side that represents the pricing decision of the firm. Third, I add a dummy version of the model to highlight mechanisms.

4.1 Demand

Consumer i's conditional indirect utility of consuming food product j of product group g in region r and week w is modeled as a linear function of product characteristics (McFadden, 1974):

$$u_{ijrw} = \alpha p_{jrw} + x_{jrw}\beta + \xi_{jrw} + \zeta_{igrw} + (1-\sigma)\varepsilon_{ijrw},$$

where p_{jrw} is the price of good j in market rw, market rw is a region in a certain week, and α is the price sensitivity. The model includes x_{jrw} as a vector of observable product characteristics of product j in market rw, with β being a vector of preferences for the observable product characteristics[13]. Unobservable product characteristics ξ_{jrw} are observable for both consumers and firms, but unobservable to the econometrician. I allow tastes for products in the same group g to be correlated, where ζ_{igrw} is a group error shock and σ is the strength of the correlation of products within a group. A group is defined as a food type (e.g. beef or chicken). The econometric error ε_{ijrw} is assumed to be iid extreme value type 1 distributed. Figure 3 shows a stylized representation of the nesting structure.

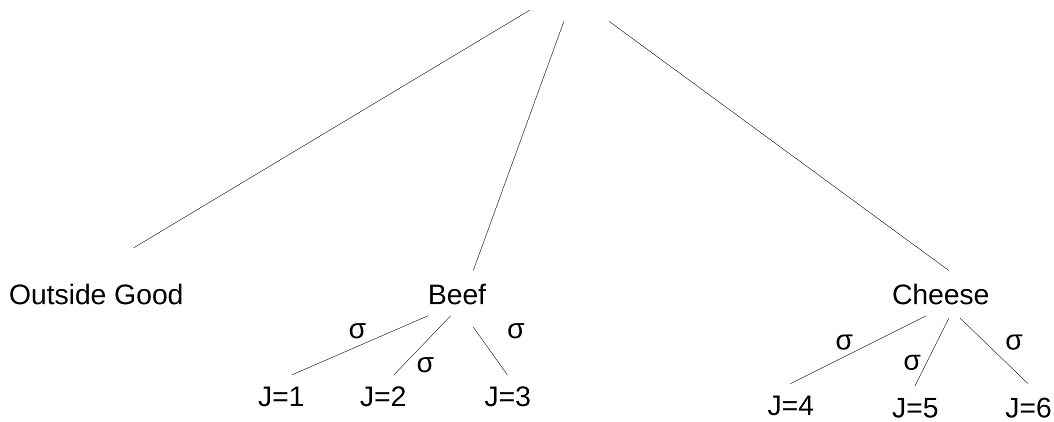
This discrete choice framework assumes that one product is being chosen among all options. Continuous or multiple-discrete models require individual-level data (Dubé, 2004; Hendel, 1999). While I cannot exclude that consumers choose several products at one occasion, multiple-discreteness is less likely to be present for products with short expiry dates. Conceptually, the utility function can be interpreted to represent utility from a future consumption occasion, with consumers making multiple decisions at each shopping trip. As I have weekly data, a purchasing occasion is considered a weekly shopping trip.

The individual utilities u_{ijrw} are aggregated over consumers i to obtain the market share of product j. Since we can only analyze differences in utility and not their absolute levels, I introduce an outside good and set its

[12] Although I also tried a random-coefficients specification, the estimated heterogeneity in price sensitivity is economically small (not reported), so I proceed with the more parsimonious specification.

[13] The observed product characteristics x_{jrw} include labels, brands, and sales variables.

Figure 3: Stylized nesting structure



Note: This figure shows the stylized nesting structure. The true nesting structure has 10 food types (poultry, pork, beef, veal, lamb, fish, cheese, eggs, legumes, and tofu), not just the two shown.

utility to 0. As described in section 2, this outside good captures the option not to buy any of the protein-rich products.

The mean utility of food product j is denoted by $\delta_{jrw} = \alpha p_{jrw} + x_{jrw}\beta + \xi_{jrw}$. Assuming that consumers choose the product that maximizes their utility, the probability that consumer i chooses product j in market rw is given by a nested logit form. Summing across consumers yields market shares for market rw: $s_{jrw} = \sum_i s_{ijrw}$.

Define the inclusive value of nest g as $D_{grw} = \sum_{k\in g} e^{\delta_{krw}/(1-\sigma)}$.

$$\begin{aligned} s_{jrw} &= \int ... \int 1(u_{ijrw} > u_{ikrw} \forall k \neq j) p(\varepsilon_{i1rw}, ..., \varepsilon_{iJrw}, \zeta_{igrw}) \\ &= \frac{1}{1+\sum_h D_{hrw}^{1-\sigma}} \text{ if } \quad j = 0 \\ &= \frac{e^{\frac{\delta_{jrw}}{1-\sigma}} D_{grw}^{-\sigma}}{1+\sum_h D_{hrw}^{1-\sigma}} \text{ if } \quad j \in J \end{aligned} \tag{1}$$

4.2 Supply

The supply side should ideally be modeled as an oligopoly where several firms compete with each other. I do not observe competing retailers, so I cannot identify an oligopoly pricing game. Instead, I summarize pricing with a fixed per-unit margin, equal to the difference between the observed price and wholesale cost in the data, and let cost and tax changes pass through to prices one-for-one[14].

For each product j in market rw I observe the baseline retail price p_{jrw} and the wholesale cost w_{jrw}, and define the baseline margin as

$$m_{jrw} = p_{jrw} - w_{jrw}.$$

In any counterfactual, the retailer holds this margin fixed and applies it to the counterfactual wholesale cost w_{jrw}^{cf}, so that the counterfactual price is

$$p_{jrw}^{cf} = w_{jrw}^{cf} + m_{jrw}. \tag{2}$$

Two features of equation 2 drive the counterfactuals. First, cost changes pass through completely: $\partial p_{jrw}^{cf}/\partial w_{jrw}^{cf} = 1$. A tariff acts multiplicatively on the duty-inclusive wholesale cost, so removing a tariff t_j lowers the cost to $w_{jrw}^{cf} = w_{jrw}/(1+t_j)$, while a per-unit Pigouvian tax τ_j raises it to $w_{jrw}^{cf} = w_{jrw} + \tau_j$. Second, policies that change only the choice set (buying local, vegetarian scenarios) leave wholesale costs unchanged, so the prices of the surviving products do not move and the entire response works through demand-side substitution.

Full pass-through of a per-unit cost change at a constant absolute margin is the standard benchmark for a large assortment retailer and is consistent with reduced form retail pass-through estimates close to one (Bonnet and Réquillart, 2013; Genakos and Pagliero, 2022; Hanson and Sullivan, 2009; Miller et al., 2017).

4.3 Intuition

To help with the intuition of the model and the interpretation of the results, this section presents a simpler version of the model. For that, I depart from the nested logit assumption and consider a logit model instead. The intuitions remain the same. For conciseness, I skip the rw market subscript. In a logit framework, the demand equation is given by:

$$s_j = \frac{e^{\delta_j}}{1 + \sum_k e^{\delta_k}} \tag{3}$$

[14]Once a consumer is in the store, the retailer is effectively a multiproduct monopolist for that shopping trip. However, the implied markups from a multiproduct monopoly do not yield reasonable results as detailed in Section 5.3.

For simplicity, I assume that $\delta_j = \alpha p_j$ and that the number of products k is 2.

The formulas for the own- and cross-price elasticities are given by:

$$\text{OPE} = \frac{\partial s_j(p_j)}{\partial p_j} \frac{p_j}{s_j} = \alpha p_j (1 - s_j)$$

$$\text{CPE} = \frac{\partial s_j(p_j)}{\partial p_l} \frac{p_l}{s_j} = -\alpha p_l s_l$$

If the price of one product increases, the quantity of that same product decreases and the quantity of the other product increases. However, in more realistic scenarios, there are several prices that change at the same time. In what follows, I refer to the contribution of a product's own price change to its quantity change as the own-price channel, and the contribution of other products' price changes as the cross-price channel. When several prices move at once, the sign of the net quantity response depends on which channel dominates. Appendix section 10.2 gives a numerical example.

Turning to the supply side, the fixed-margin rule of equation 2 keeps the intuition simple. Each product's price is its wholesale cost plus a constant margin, $p_j = w_j + m_j$, so a cost change moves the price one for one while a change in the choice set leaves the surviving products' prices untouched. This adds a third channel alongside the own- and cross-price channels: a cost channel that operates only in scenarios changing wholesale costs (tariff liberalization, a Pigouvian tax). In pure choice-set scenarios (buying local, removing beef, vegetarianism) there is no cost change, prices do not move, and the quantity response is governed entirely by the own- and cross-price elasticities.

5 Identification and estimation

I estimate the demand parameters by two-stage least squares, instrumenting for the two endogenous regressors. The supply side requires no estimation: the per-unit margin in equation 2 is read directly from the baseline data. This section describes the demand estimation and reports the results.

5.1 Demand estimation

Estimating the market shares s_{jrw} is complicated by the fact that the econometric error enters nonlinearly. I solve for the mean utility δ_{jrw} by inverting the market shares and estimate the following linear equation (Berry, 1994):

$$ln \frac{s_{jrw}}{s_{0rw}} = \delta_{jrw} + \sigma ln \frac{s_{jrw}}{s_{grw}} = \alpha p_{jrw} + x_{jrw}\beta + \sigma ln \frac{s_{jrw}}{s_{grw}} + \xi_{jrw}$$

where $\frac{s_{jrw}}{s_{grw}}$ is the market share of product j within food type g in market rw and the unobserved product characteristic ξ_{jrw} is the error term. The observed product characteristics x_{jrw} include labels, brands, and sales variables. I refer to these variables as "controls" in the next equation. Table 3 shows the median market shares for the median domestic and imported food type.

Table 3: Market Shares Summary Statistics

Food Type	Imported	Market Share	Group Market Share	Within Group Market Share
Beef	0	1.21	1.21	99.70
Beef	1	0.00	1.21	0.30
Cheese	0	4.94	5.34	92.40
Cheese	1	0.41	5.34	7.60
Eggs	0	1.53	1.85	82.88
Eggs	1	0.32	1.85	17.12
Fish	0	0.42	0.42	98.66
Fish	1	0.01	0.42	1.34
Lamb	0	0.08	0.11	74.40
Lamb	1	0.03	0.11	25.60
Legumes	0	0.29	0.37	77.42
Legumes	1	0.08	0.37	22.58
Pork	0	2.98	2.98	99.90
Pork	1	0.00	2.98	0.10
Poultry	0	2.57	2.60	98.68
Poultry	1	0.03	2.60	1.32
Tofu	0	0.14	0.18	80.96
Tofu	1	0.03	0.18	19.04
Veal	0	0.28	0.29	98.58
Veal	1	0.00	0.29	1.42
Outside Good	0	84.65	100.00	100.00

Note: Median market shares (in %) across markets, by food type and import status (Imported = 1 for imported products, = 0 for domestic). 'Market Share' is the cell's share of the entire market, including the outside good; 'Group Market Share' is the food type's combined share (domestic plus imported); 'Within Group Market Share' is the cell's share within its food-type nest, so the two import-status rows within a food type sum to 100. The final row is the outside good (all other food).

The parameters α and σ might both suffer from endogeneity which can be solved using an instrumental variable approach. First, I explain the types of biases that arise for each parameter. Second, I propose instrumental variables to obtain unbiased estimates of α and σ. The estimating equation for the demand estimation is:

$$ln\frac{s_{jrw}}{s_{0rw}} = \alpha p_{jrw} + \sigma ln\frac{s_{jrw}}{s_{grw}} + \theta_g + \theta_{year} + \theta_{quarter} + \theta_r + controls + \xi_{jrw},$$

where $\frac{s_{jrw}}{s_{0rw}}$ is the relative market share of product j. p_{jrw} is the price of good j in region r in week w. $\frac{s_{jrw}}{s_{grw}}$ is the within group market share, where a group is defined as a food type. This is the variable that makes this a nested logit, excluding it would leave us with a logit specification. θ_g, θ_{year}, $\theta_{quarter}$, and θ_r are group, year, quarter, and region fixed effects. The control variables include labels, brands, and sales variables.

The main variable of interest is the price coefficient α. This coefficient might be upward biased, since higher values of the unobserved characteristics in the error term might be associated with a higher price. Including fixed effects that vary by region, year, and quarter help to control for demand shifts caused by spatial components, yearly trends, and seasonal variation. The within group market share $\frac{s_{jrw}}{s_{grw}}$ measures the market share of e.g. domestic beef with respect to the market share of all beef products. It is correlated with the unobserved product characteristic ξ_{jrw} in the error and thus endogenous. Products with more desirable characteristics might have a higher market share within their group.

A valid instrument for the price should be relevant and exogenous. I use the wholesale cost for each product as an instrument for the price. This instrument is relevant because it is part of the price. The exogeneity condition holds if the instrument is uncorrelated with the error term. Wholesale costs as measured by the European retailer represent production costs and are not sensitive to particular demand fluctuations. For the subset of products where I observe the internal cost calculation, the raw material accounts for the large majority of the wholesale cost, with logistics, transport, and tariffs together a small residual. This supports treating the wholesale cost as a commodity-driven cost shifter: the components one might worry are correlated with origin-specific quality or freshness are a minor part of the cost, while the dominant component is the upstream commodity price, which is set in wholesale markets and orthogonal to consumer-level demand shocks conditional on the time fixed effects.

To instrument for the within-group market share, I construct a proxy for the advertising exposure of other products in the same nest — that is, other varieties within the retailer's own assortment (e.g. other beef products). Specifically, I use the average share of revenue from planned sales of products $k \neq j$ within group g, in region r, and week w. The exclusion restriction is that planned advertising for products $k \neq j$ affects j's demand only through within-group substitution. The behavioral logic is that a sale banner on another beef product draws attention within the beef set, shifting which beef variety a consumer chooses rather than changing whether the consumer buys beef at all, conditional on prices and j's own promotion. Importantly, the price effect of these sales is fully captured in the price variable; the instrument is meant to isolate the advertising or visibility effect.

Because sales are fixed months ahead on promotional calendars while demand is observed weekly, the instrument is predetermined with respect to the weekly demand shocks ξ_{jrw} it must be orthogonal to. I exclude expiry sales, which are often reactive to low demand. I argue that planned sales of competing products are likely exogenous to the unobserved demand for product j, conditional on controls.

The relevance of the instrument comes from the fact that increased advertising of other products in the group reduces product j's within-group market share. The exclusion restriction is that planned advertising for product $k \neq j$ affects product j's demand only through substitution within the group, not through direct utility effects. In practice, I compute:

$$proxy\ for\ sales = mean_{rw,k\neq j\in g}(sales)$$

I calculate the instrument as the mean percentage of revenue from products on sale in market rw for food type g, where I exclude product j from the calculation. I use the standard sales for creating the instrument as opposed to packaged sales or expiry sales. Using packaged sales leads to similar results. Expiry sales are not planned ahead and are likely to be correlated with demand. An increase in expiry sales means that consumers buy less of a product than expected. Appendix Figure 15 shows the variation in the instrument and the within group market share. The correlation is negative: When the instrument increases, the within group market share decreases. This happens because for higher levels of the instrument, other products in the same group become more attractive, thus the within group market share of product j decreases.

5.2 Regression results

This section shows the regression results for the nested logit model and the corresponding elasticities. In Table 4, I report the nested logit demand estimates. Column 1 shows an OLS specification without instruments, column 2 shows the IV specification that instruments for price with wholesale costs and for the within-group share with the advertising proxy, and columns 3 and 4 show the corresponding first stages. All columns include continent, food type, quarter, year, and regional fixed effects, and a set of controls.[15] The control variables include the percentage of products with a brand name (e.g. "European Retailer Premium Brand") and the percentage of products with different types of labels (e.g. "organic"). Most labels and brands are omitted due to space constraints, but shown in Appendix Table 11. The sales variables represent different types of sales and enter with differing signs. As noted before, the price effect of sales is captured by the price variable. Therefore, all three sales variables represent the effects of advertisements related to sales, e.g. special shelf placement or stickers on products. The standard sales (sales 1) and packaged sales (sales 2) variables combine price reductions with advertisements. Thus, a higher share of sales increases the market share of a product. The expiry sales (sales 3) represents sales due to a product being close to its expiry date. The retailer only uses this type of sales if consumers purchased less of a product than expected. Therefore, a higher share of expiry sales indicates lower demand than expected and thus decreases the relative market share.

My preferred estimates come from the IV specification in column 2. Instrumenting moves the price coefficient away from zero relative to OLS, confirming that the OLS coefficient is biased upward by unobserved quality, and yields a price coefficient of $\alpha = -0.016$. The coefficient on the within-group share is $\sigma = 0.621$, which lies in the unit interval as required and implies that varieties within a food type substitute more readily with one another than across food types.

[15] Interacting the fixed effects or including week-of-the-year fixed effects does not alter the results.

Table 4: Nested Logit Regressions

	Dependent variable:			
	relative MS NL	relative MS NL IV	price FS 1	within group MS FS 2
	(1)	(2)	(3)	(4)
Price	-0.004^{***}	-0.016^{***}		
	(0.0002)	(0.001)		
Within Group MS	0.935^{***}	0.621^{***}		
	(0.002)	(0.014)		
Sales 1	0.797^{***}	2.133^{***}	-15.315^{***}	4.809^{***}
	(0.023)	(0.066)	(0.540)	(0.078)
Sales 2	-0.500^{***}	-1.999^{***}	-24.475^{***}	-3.796^{***}
	(0.024)	(0.074)	(0.544)	(0.078)
Sales 3	0.282^{***}	0.674^{***}	-7.961^{***}	1.531^{***}
	(0.009)	(0.021)	(0.210)	(0.030)
Label Organic	-0.205^{***}	-0.132^{***}	10.833^{***}	-0.085
	(0.016)	(0.024)	(0.394)	(0.057)
Label Local	0.057^{***}	0.394^{***}	-8.741^{***}	1.272^{***}
	(0.013)	(0.023)	(0.309)	(0.045)
Wholesale Cost			0.900^{***}	-0.038^{***}
			(0.004)	(0.001)
Advertising Proxy			-4.478^{***}	-3.379^{***}
			(0.816)	(0.118)
F-Statistic			4954	1282
Observations	39,262	39,262	39,262	39,262

Note: *p<0.1; **p<0.05; ***p<0.01

Column (1) is the nested logit estimated by OLS and column (2) by instrumental variables. The dependent variable in both is the log relative market share (log product share minus log outside-good share). Columns (3) and (4) are the corresponding first stages, for price and for the within-group market share respectively, with wholesale cost and the advertising proxy as the excluded instruments. All specifications include quarter, year, region, continent, and food-type fixed effects, plus label and brand controls.

Columns 3 and 4 show the first stage estimates for the nested logit regression. Increasing wholesale costs by 1 € results in a consumer price increase of around 0.90 € in column 3. For the within group market share first stage in column 4, the proxy for advertisement enters negatively. Increasing the proxy of advertisement of other products increases the denominator of the regressor and thus decreases the within group market share. In other words, the group becomes more desirable, so the within group market share of product j decreases. The first stages are strong with high F-statistics.

The label variable captures the additional preference for a visible label certifying domestic provenance, identified separately from the Home fixed effect that already absorbs domestic origin. In the IV specification (column 2), the label coefficient of 0.394 implies that moving a product from no local labeling to fully labeled raises its market share relative to the outside good by about 48% $[e^{\beta_{\text{label}}} - 1]$.

This specification accounts for consumer preferences for more environmentally friendly products as long as these preferences are captured by food type and continent fixed effects. Alternatively, the greenhouse gas emissions intensities for each product could enter the utility function. As emissions intensities vary across continents and across products, my setup does not allow to estimate a coefficient on emissions intensity. A specification in Appendix Table 13 where instead of continent fixed effects I only control for domestic or imported products and include greenhouse gas emissions intensity shows that the coefficients of interest do not change and that the coefficient on emissions intensity is very low. As consumers do not observe emissions intensities, but rather product characteristics like food type and country of origin, I do not include emissions intensities in the main specification.

All demand regressions in the main text use the aggregated data. As mentioned in section 2, the level of aggregation is dictated by the emissions data: greenhouse gas intensities are available at the level of broad product categories rather than individual barcodes, so the counterfactual emissions calculations that are the focus of this paper can only be carried out at the category level. The aggregated demand estimates are the relevant inputs to those calculations.

To verify that the demand structure is not an artifact of aggregation, Appendix Table 12 reports the same specification estimated on the barcode-level data, in OLS and IV form and with and without product fixed effects. The estimates are consistent with the aggregated results. In the instrumented specification without product fixed effects, which is the direct analogue of the main specification, the within-group coefficient is 0.39 (versus 0.62 on aggregated data) and the price coefficient is -0.013 (versus -0.016). In both cases instrumenting reduces the within-group coefficient substantially relative to OLS, where it approaches one.

The own- and cross-price elasticities are given in Table 5. In the case of several products (several continents to import from or several transport categories), I present the median elasticity of the group. The own-price elasticities range between -0.07 and -1.5, with larger elasticities for ruminants like beef or lamb. These magnitudes are well within the range reported for meat in the previous literature (Anders and Mőser, 2010;

Eales and Unnevehr, 1988; Reed et al., 2003; Yang et al., 2018), and the pattern of beef and other ruminants being the most elastic products matches earlier estimates (Lanfranco and Rava, 2014; Lusk and Tonsor, 2016).

The second column shows the cross-price elasticities for products within the same group, e.g. between domestic and imported beef. The third column shows the cross-price elasticities for products outside the group, e.g. between beef and chicken. Allowing for higher cross-price elasticities within groups is a feature of the nesting structure. Without nesting, all cross-price elasticities would take the logit form of column 3. The reason for these low cross-price elasticities lies in the data structure: low market shares for the imported products yield low cross-price elasticities. I report the formulas for calculating the elasticities in a nested logit setup in Appendix section 10.4. The share switched to the outside good in the fourth column ranges from 6%-16%. This is the share of the substitution away from product j that goes to the outside good when its price rises, rather than to other protein-rich products (Berry et al., 1995).

Table 5: Nested Logit IV Elasticities

Group	OPE NL	CPE Within Group	CPE Across Groups	Share switched to OG (%)
Beef Dom	-0.4282	0.554882	0.004475	13.9
Beef Imp	-2.9478	0.004977	0.000041	6.0
Cheese Dom	-0.1570	0.226538	0.007170	14.7
Cheese Imp	-0.4226	0.021079	0.000667	6.3
Eggs Dom	-0.0847	0.085998	0.001004	11.9
Eggs Imp	-0.1239	0.013722	0.000168	6.8
Fish Dom	-0.5067	0.379466	0.001435	10.3
Fish Imp	-1.0915	0.008185	0.000030	6.3
Lamb Dom	-0.5752	0.234161	0.000259	8.6
Lamb Imp	-1.6924	0.154087	0.000196	6.8
Legumes Dom	-0.0392	0.035392	0.000082	11.5
Legumes Imp	-0.0647	0.010659	0.000024	6.9
Pork Dom	-0.3001	0.512652	0.008967	16.7
Pork Imp	-2.1618	0.001155	0.000022	6.1
Poultry Dom	-0.2435	0.392735	0.006394	15.5
Poultry Imp	-0.7954	0.007562	0.000103	6.1
Tofu Dom	-0.1026	0.101510	0.000110	11.9
Tofu Imp	-0.1548	0.023114	0.000023	6.8
Veal Dom	-0.4311	0.667277	0.001077	17.3
Veal Imp	-2.4075	0.021789	0.000037	6.8

Note: This table shows the own- and cross-price elasticities of the IV nested logit specification. 'OPE' stands for own price elasticity CPE Within Group' for cross-price elasticity within groups (e.g. from imported to domestic beef), 'CPE Across Groups' for cross-price elasticity across groups (e.g. from beef to chicken). The 'share switched to OG' refers to the share of expenditure that switches to the outside good, relative to all expenditure switching away from product j in response to a small increase in its price.

A natural point of comparison is the quantitative trade-and-climate literature, which calibrates substitution using a nested CES structure (Table 6), with an across- to within-group substitution ratio in the range of roughly 0.52 to 0.72 (Correa-Dias et al., 2026; Costinot et al., 2016; Farrokhi and Pellegrina, 2023). A single directly comparable ratio cannot be read off the two frameworks, because the within- and across-group cross-price elasticities are evaluated at different share objects. What the estimates deliver instead is an unambiguous qualitative message: substitution across food groups is far weaker than substitution within them, implying a steeper hierarchy than these calibrations impose. Therefore, models that adopt values in the 0.52–0.72 range understate consumers' resistance to cross-group substitution.

The elasticities make this concrete: for domestic beef, for example, the within-nest cross-price elasticity is on the order of 50 times larger than the across-nest one, so domestic and imported beef are far more substitutable than domestic beef and domestic chicken. This pattern holds across product categories and is exactly the structure a nested model is built to capture. It reinforces the value of estimating these substitution patterns directly rather than calibrating them: the data point to a substitution hierarchy at least as steep as, and plausibly steeper than, the one built into current trade-and-climate models.

Table 6: Comparison with the quantitative general equilibrium literature

Quantity	This paper	Correa-Dias et al.	Farrokhi & Pellegrina	Costinot et al.
Within-group substitution	–	5.00	5.76	5.40
Across-group substitution	–	3.50	4.16	2.82
Across/within ratio	**0.003**	**0.70**	**0.72**	**0.52**

Note: Columns report each general equilibrium paper's calibrated CES elasticities of substitution within and across groups, together with the implied across/within ratio. The nested logit estimates of this paper are not comparable in levels to these CES parameters, and a single across/within ratio cannot be cleanly recovered because the within- and across-group cross-price elasticities are evaluated at different share objects. As discussed in the text, the estimates nonetheless imply a steeper substitution hierarchy than any of these calibrations.

5.3 Supply-side conduct

Ideally, the supply side would be modeled as Bertrand–Nash price competition among the retailers active in Home. I observe only one retailer, so I cannot identify a multi-firm pricing game. Instead, I could treat the retailer as a multiproduct monopolist over each shopping trip and recover markups from the pricing first-order conditions, $p_{rw} = mc_{rw} + \Omega_{rw}^{-1} s_{rw}$, where Ω_{rw} collects the own- and cross-price derivatives of shares. This is not viable in my setting. The market is large and fragmented, so individual product shares are small, Ω_{rw} is close to singular, and the implied markups explode: the associated Lerner indices exceed one for [68.5]% of products, which is implausible for a grocery retailer. Additionally, the markup term is essentially invariant to the nesting parameter σ across its admissible range, so a jointly estimated demand and supply GMM system with a monopoly pricing moment adds almost no information about σ while importing the near-singularity of Ω_{rw}^{-1} into the demand estimates. Alternative conduct assumptions do not resolve this: a single-product monopoly markup produces the same above-one Lerner indices, and joint estimation under either conduct returns demand parameters that are unstable and driven by the matrix inversion rather than by the data.

For these reasons I summarize the retailer's pricing with a fixed per-unit margin, $m_{jrw} = p_{jrw} - w_{jrw}$, read directly from the baseline data and held constant in counterfactuals, so that cost and tax changes pass through one-for-one (equation 2). Full pass-through at a constant absolute margin is the standard benchmark for a large-assortment retailer and is consistent with reduced-form retail pass-through estimates close to one (Bonnet and Réquillart, 2013; Genakos and Pagliero, 2022; Hanson and Sullivan, 2009; Miller et al., 2017). This choice is also conservative for the paper's main message: it shuts down any competitive price response, so the choice-set counterfactuals (buying local, vegetarianism) operate purely through demand-side substitution and do not rely on a conduct assumption at all. Only the cost-based scenarios (tariff liberalization and the Pigouvian taxes) depend on pass-through, and there the one-for-one benchmark is well supported.

6 Counterfactuals

The previous sections outlined a discrete choice model for meat demand, discussed identification and presented the regressions results. In this section, I use the estimates of the model to analyze some counterfactual scenarios. These hypothetical scenarios rely on changing the choice set or the prices and comparing the outcome to the status quo. I introduce three sets of scenarios: the first considers trade-related scenarios like buy local (or autarky) and free trade. The second category revolves around reducing meat consumption like vegetarianism or a no-beef no-cheese scenario. The third set of scenarios implements Pigouvian taxes, so taxes that are proportional to the emissions intensity of each product.

Buying local and vegetarianism are popular, supposedly eco-friendly strategies, as evidenced by a representative survey of food consumption and sustainability (BEUC, 2020). Often, counterfactuals in economic models are perceived as policies imposed by the government. However, they can also be viewed as self-imposed restrictions. In this context, the vegetarian scenario would correspond to all consumers voluntarily imposing mental taxes on meat. While some of these counterfactuals may seem extreme, they illustrate the potential environmental impact if these popular movements were to be fully embraced.[16] Conversely, the conventional economic approach is to account for the externality from emissions and impose a Pigouvian tax.

I structure the analysis in five subsections. First, I explain how the counterfactuals are calculated. Second, I show the status quo of food consumption and the related total emissions. Third, I discuss several trade-related policies, revolving around buy local (or autarky) and free trade. Fourth, I analyze policies related to reducing meat consumption, including a no beef, a vegetarian, and a no beef no cheese scenario. Fifth, I discuss three Pigouvian tax scenarios.

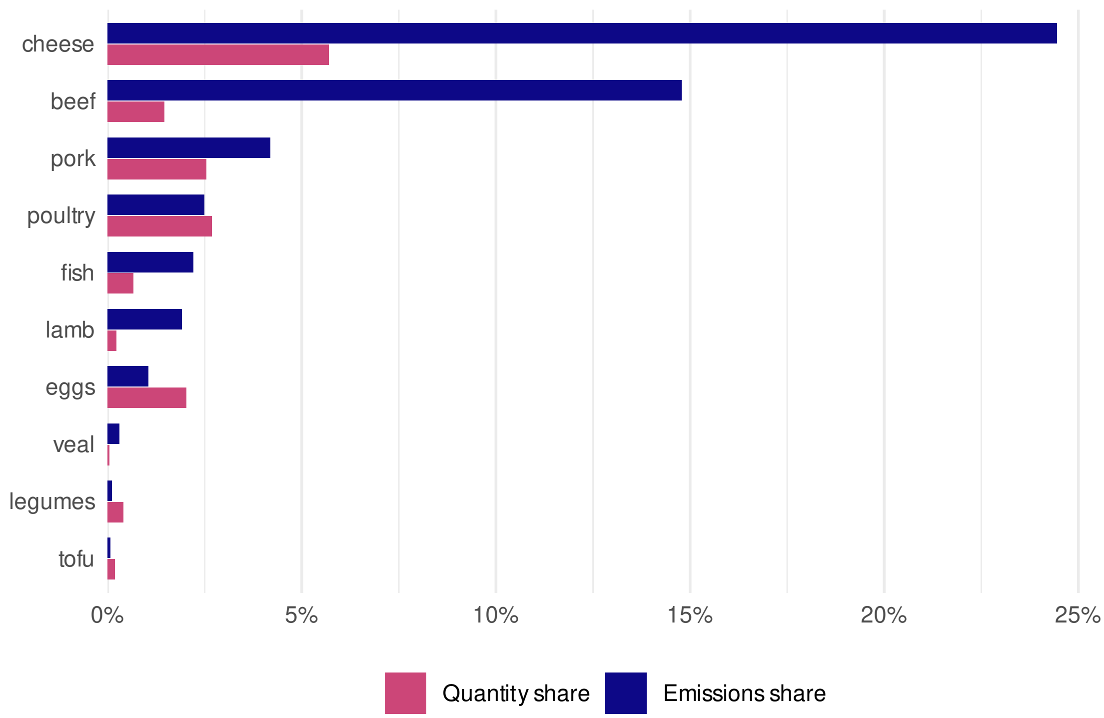
6.1 Status quo

Figure 4 shows the current consumption patterns in Home. Beef and cheese are the highest contributors to emissions. While beef only represents around 1.5% of consumption in terms of quantity, it is responsible for

[16] Calculating extreme counterfactuals stretches the interpretation of the elasticities that only hold for local changes.

around 15% of emissions. The protein-rich products considered in my sample account for only 16% of total consumption in terms of quantities while yielding more than 50% of emissions (see Appendix Table 14). I compare all counterfactual scenarios to this status quo. As the discrete choice model compares quantities relatively to an outside good, the total amount of food consumed is held fixed across counterfactuals.

Figure 4: Quantity vs. emissions share by food type in the status quo



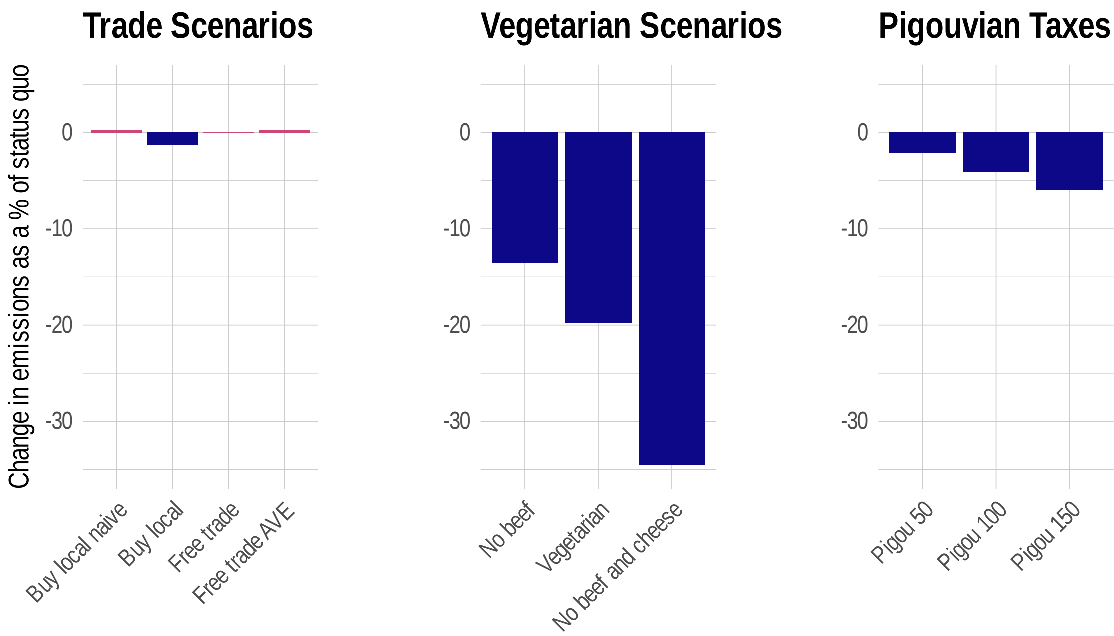
Note: Each food type has two bars: its share of total consumption (by quantity) and its share of total emissions. Both totals include the outside good (all other food), which is itself omitted from display, so the bars do not sum to one.

6.2 Overview of counterfactuals

To run counterfactuals, I compute how prices and quantities change under each policy. On the supply side, I assume the retailer holds its per-unit margin fixed, so cost and tax changes pass through to prices one-for-one: the counterfactual price of a product is its counterfactual marginal cost plus its observed baseline margin. On the demand side, consumers reallocate across products according to the estimated demand system. Concretely, for each scenario I construct the counterfactual marginal cost adjusting for the tariff or tax where relevant, or leaving it unchanged where the policy only alters the choice set. I add the fixed margin to obtain the counterfactual price, and then evaluate the demand system (equation 1) at these prices over the counterfactual set of products to obtain the new market shares and quantities.

Figure 5 summarizes the counterfactuals, showing the average yearly change in emissions relative to the status quo for each scenario. Three patterns stand out. First, the trade-related scenarios move emissions only slightly, and the buy-local case flips sign once substitution is estimated rather than assumed: the naive accounting scenario raises emissions, while the structural scenario lowers them. Second, the diet-based scenarios deliver by far the largest reductions, even though the no beef and no beef no cheese scenarios leave every other meat type in the choice set. Third, Pigouvian taxes reduce emissions across the board while preserving the full choice set, with reductions that scale in the tax rate. The remainder of this section takes each scenario in turn.

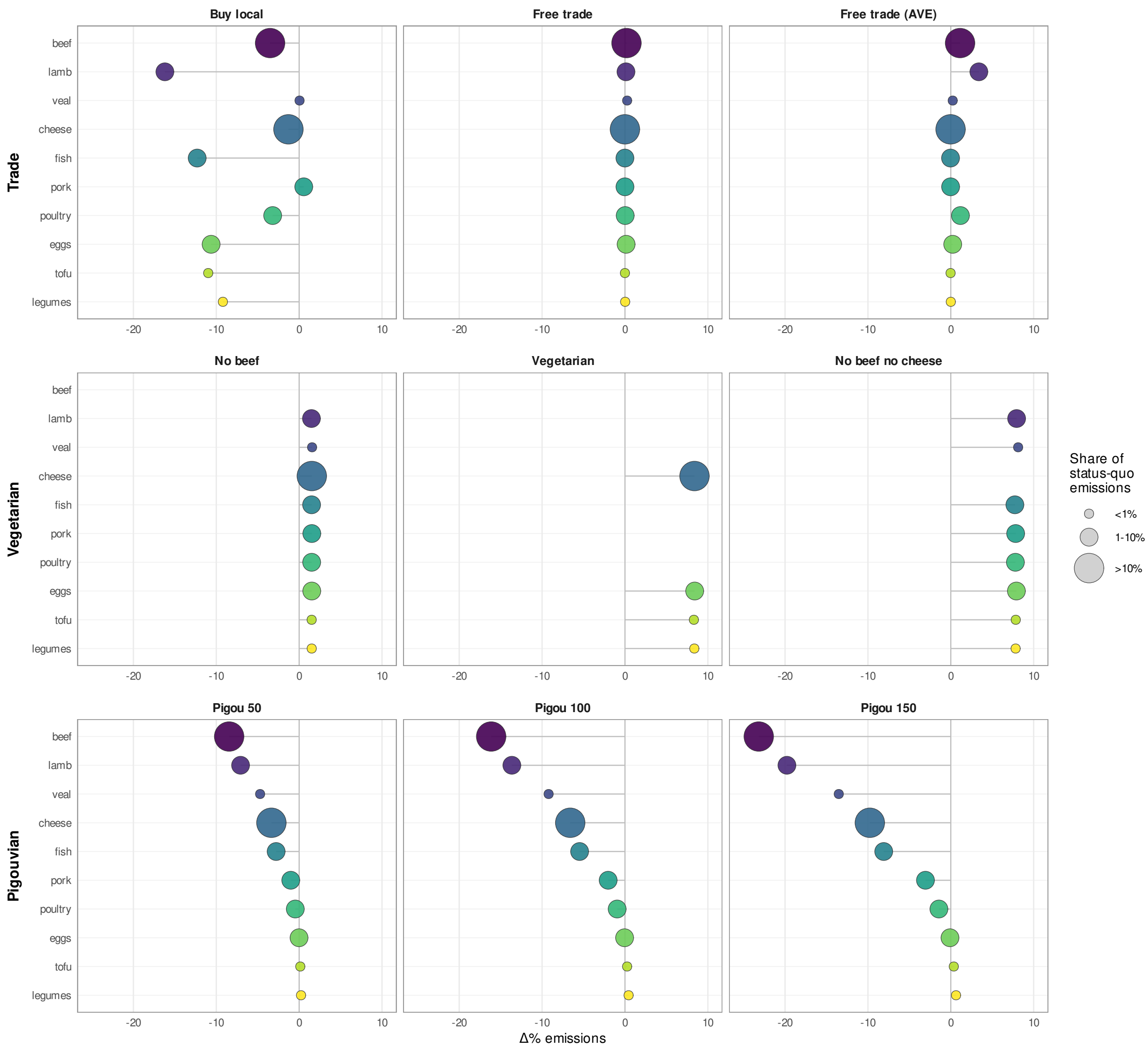
Figure 5: Summary of counterfactuals



Note: Each bar is a counterfactual scenario. The y-axis is the resulting change in total GHG emissions, expressed as a % of status-quo emissions, for an average year. Panels group the scenarios into trade, vegetarian, and Pigouvian tax policies.

To illustrate the mechanisms for each counterfactual, Figure 6 shows the change in emissions for each food type compared to the status quo. The top three graphs are the trade-related scenarios, the middle ones the vegetarian ones, and the last three show the Pigouvian taxes. The size of the markers corresponds to the share of emissions in the status quo.

Figure 6: Emissions response by food type across counterfactual scenarios



Note: Each point is a food type. Food types and colors are ordered by emissions intensity (kgCO_2e per kg), and marker size shows the food type's share of total status-quo emissions. Products excluded by a scenario are omitted from that panel. The x-axis shows the change in emissions compared to the status quo for an average year.

6.3 Trade-related counterfactuals

6.3.1 Buy local

The buying local - or autarky - scenarios reflect a situation in which no protein-rich imported products are available for purchase[17]. Alternatively, this scenario can be interpreted as a complete adherence to the buy local idea in the form of "mental tariffs." In any case, consumers do not purchase imported protein-rich foods in autarky. I provide three versions of the buy local scenario: The first is a naive, accounting-type scenario in which I do not make use of the estimated elasticities. Instead, I assume that people consume the same of each food type, but that all food is produced in Home. The second counterfactual departs from this strong assumption and uses the elasticities that are estimated with the demand-side elasticities.

Figure 7 summarizes the results from this exercise: Contrary to popular belief, emissions increase in the naive accounting-style scenario. Once the substitution away from imported products is governed by demand elasticities, the sign flips and the emissions decrease. This change in the sign shows why estimating the elasticities is important. It determines whether a policy will hurt or help the environment.

The reason that the emissions increase in the naive scenario relates to the production emissions in different countries. For most food types, the emissions decrease (see Appendix Table 15)[18]. For those products, this implies that Home is ecologically more efficient than the producers of imported products. This is however not true for cheese and lamb, for example. For those two food types, the emissions increase when completely produced in Home. Overall, this leads to a slight increase in emissions. Similarly to the naive buy local scenario, Correa-Dias et al. (2026) find an increase in emissions from a scenario with trade restrictions in their general equilibrium framework that focuses on modeling food supply.

Turning to the buy local scenario with the estimated elasticities, some products experience an increase in their consumption after autarky, while others are consumed less. Figure 6 shows the resulting changes in emissions. The largest decrease in emissions happens for low-emissions products and lamb. Appendix Table 16 shows that most products are consumed less, while veal and pork are consumed more. Overall, this leads to a decrease in emissions of around 1.3%.

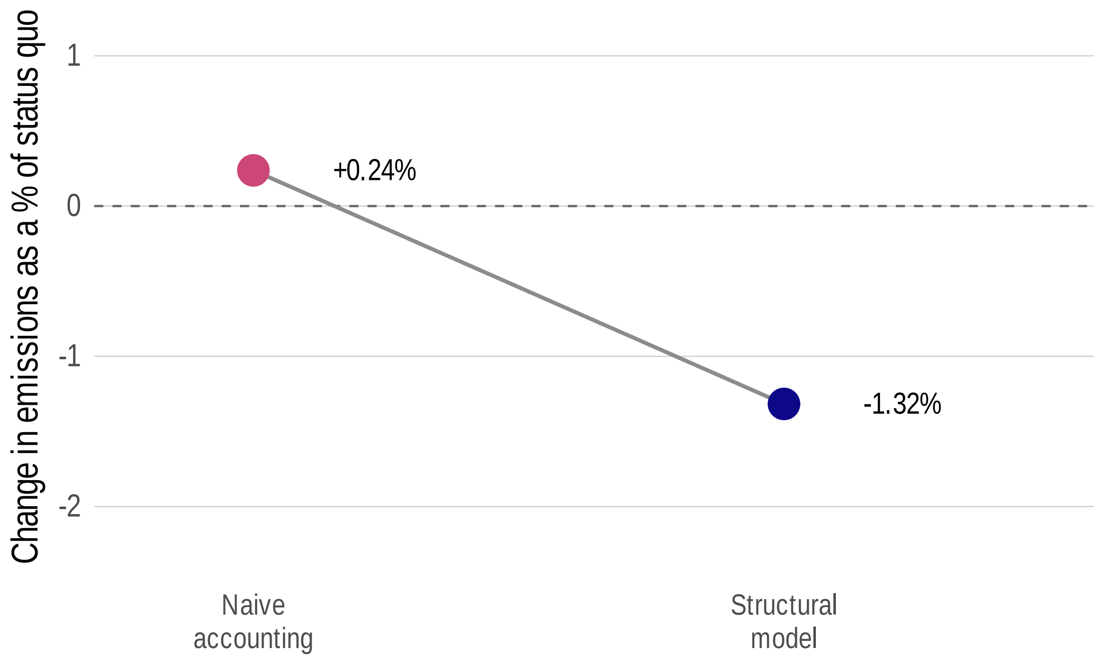
6.3.2 Free trade

The previous section considered buy local or autarky. This section deals with the opposite scenario: free trade. Analyzing a free trade scenario involves determining the tariffs for each imported product. The applied tariffs are ad valorem rates, and for imported products the observed wholesale cost w_{jrw} includes the duty the retailer pays at the border. Removing a tariff therefore lowers the wholesale cost: the duty-free wholesale cost

[17] Discrete choice models express quantities in terms of an outside good. Consequently, my autarky counterfactuals focus on protein-rich products while assuming no change in the outside good. Thus, the import mix for the outside good remains constant, even in an autarky scenario.

[18] All counterfactual tables compare the scenario ('After') to the status quo ('Before'), reporting absolute quantities in metric tons, emissions in tCO_2e, and percentage change in the $\Delta\%$ columns.

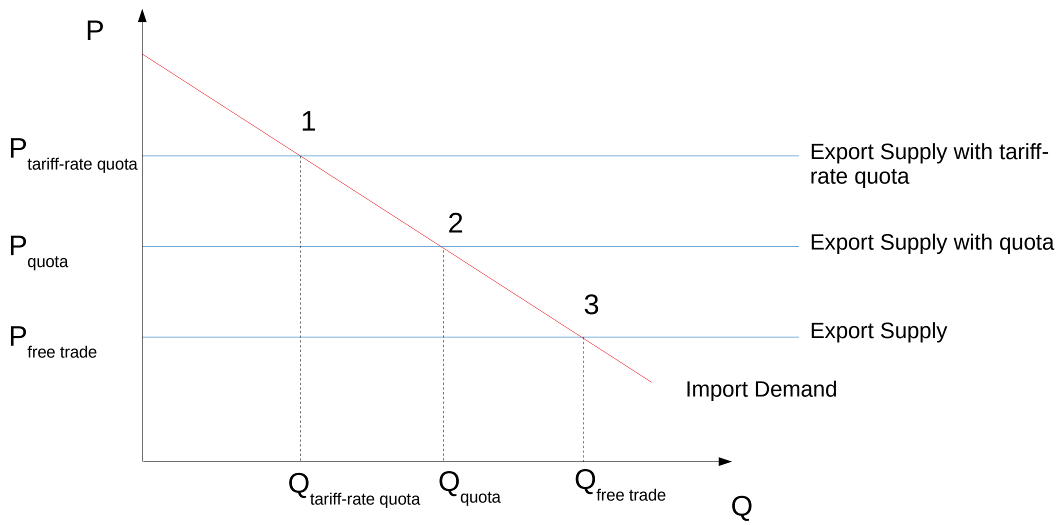
Figure 7: Buy Local Specifications



Note: The y-axis is the change in total GHG emissions under buy local, expressed as a % of status-quo emissions. The 'naive accounting' point holds each food type's consumption share fixed and assumes imported quantities are replaced one-for-one by domestic production. The 'structural model' point uses the estimated demand system, letting consumers substitute across products once imports leave the choice set.

is $w_{jrw}^{FT} = w_{jrw}/(1+t_j)$, where t_j is the ad valorem tariff. Since the retailer holds its margin fixed, this cost reduction passes through one-for-one to the consumer price: the free-trade price is $p_{jrw}^{FT} = w_{jrw}^{FT} + \text{margin}_{jrw}$, where $\text{margin}_{jrw} = p_{jrw} - w_{jrw}$ is the observed baseline margin. I assume that Home is a small open economy and takes prices as given. There are three types of trade regimes for the products in my data set: no restriction, tariff, and tariff-rate quotas. Figure 8 illustrates these different tariff regimes.

Figure 8: Prices under different tariff regimes



Note: A stylized diagram of the market for an imported product: the x-axis is quantity, the y-axis is price. The downward-sloping line is import demand; the three upward-sloping schedules are the exporter's supply under free trade, under a binding quota, and under a tariff-rate quota (TRQ). The numbered points are the equilibria under the three regimes, with the associated quantities (Q_{quota}, Q_{TRQ}, $Q_{free\ trade}$) and prices (P_{quota}, P_{TRQ}, $P_{free\ trade}$) marked on the axes.

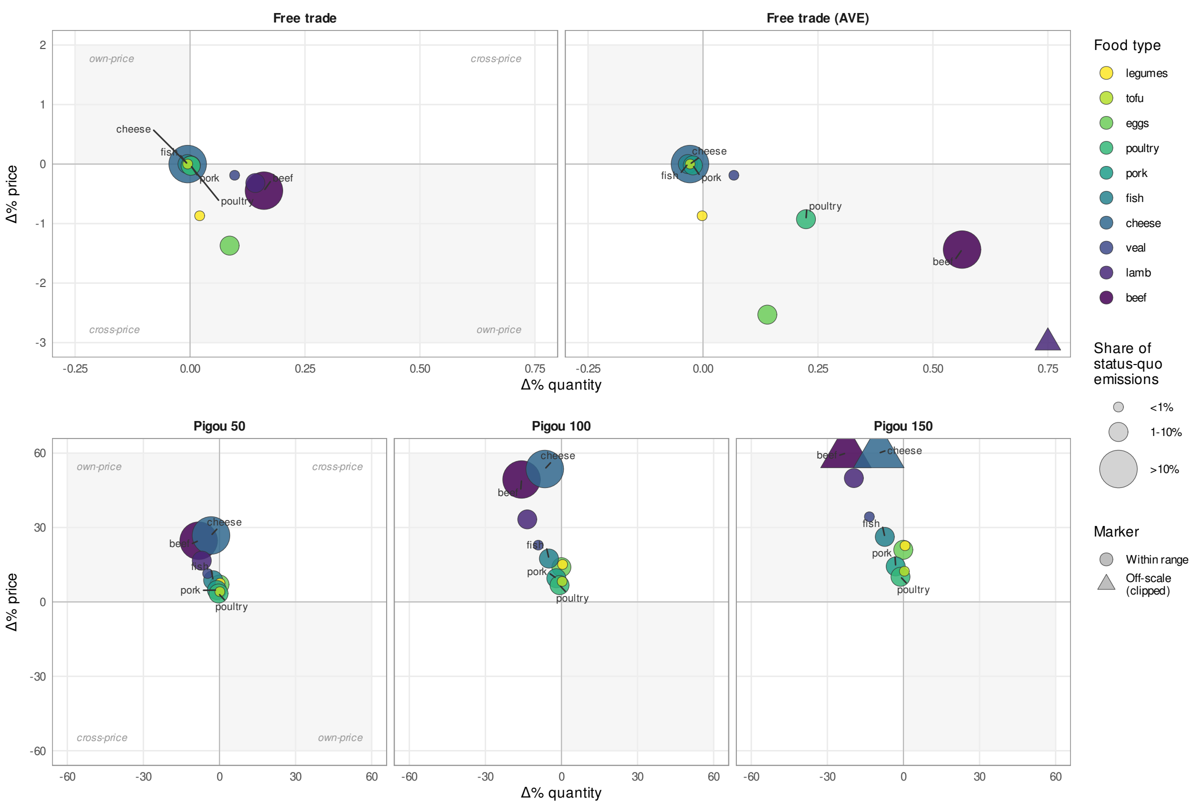
The small open economy assumption implies that the price of products without a tariff is a world price and would not change under free trade (point 3 in Figure 8). For products with a tariff, the solution is simple: One can just use the existing tariffs for each product. A subset of products is subject to tariff-rate quotas (point 1 in Figure 8). The quantity that can be imported is restricted by a quota and combined with a tariff. The tariffs on within-quota quantities are relatively low, while the tariffs for quantities outside of the quota are very high. If the quota is binding, all imports would occur within the quota. Simply subtracting the tariff would yield point 2 in Figure 8 and not the free trade price in point 3.

Because some products are subject to tariff-rate quotas, I provide two scenarios. In the first, I remove only the low within-quota tariff while the quota remains in place. Since the quota continues to bind, its rent is retained in the domestic price, so this scenario understates the price change and serves as a conservative lower bound. In the second, I remove the full ad valorem equivalent (AVE) tariff from the Market Access

Map of the World Trade Organization.

The within-quota free trade scenario leads to a modest increase in emissions of around 0.03%. The top-middle panel of Figure 6 shows that the changes in emissions are small for all food types. The reason for the increase in emissions is that both beef and veal are consumed more, as shown in Appendix Table 17. The top-right panel of Figure 6 shows the out-of-quota Free Trade scenario. Here, the emissions from beef, veal and poultry increase slightly, leading to a total increase in emissions of 0.24% (Appendix Table 17).

Figure 9: Quantity and price response by food type across counterfactual scenarios with price changes



Note: The x-axis shows the change in quantity, the y-axis the change in price. Each point is a food type. Colors are ordered by emissions intensity (kgCO_2e per kg), and marker size shows the food type's share of total status-quo emissions. Triangles mark points whose true value lies beyond the plotted range and are placed at the edge. The four quadrants correspond to different mechanisms from Section 4.3: upper-right ($p\uparrow$, $q\uparrow$) and lower-left ($p\downarrow$, $q\downarrow$) are cases in which the cross-price channel dominates—consumers substitute toward (away from) this product because other prices rose (fell) by more. Upper-left ($p\uparrow$, $q\downarrow$) and lower-right ($p\downarrow$, $q\uparrow$), shaded in light grey, are cases in which quantity moves in the direction implied by the own-price effect. A handful of products register price and quantity changes too small to resolve visually but sit in the cross-price quadrant: under free trade, legumes and pork (both quantity and price up); under the Pigouvian scenarios, tofu and legumes. Exact values are reported in the Appendix Tables.

The free trade scenarios operate through a price change: Imported products that were under a trade policy regime become cheaper and will be consumed more. The upper two panels of Figure 9 show the change in quantity and price for the trade scenarios. For the free trade scenario, the own-price channel (described in Section 4.3) dominates: The products become cheaper, so people buy more. For the AVE Free Trade scenario, this is true for most products as well. However, legumes and pork experience a decrease in price and a simultaneous decrease in quantity[19]. The reason is the cross-price channel: Although their prices decrease and people should buy more, other product's prices decrease more, so people buy less of pork and legumes and more of the other products. This cross-price channel dominates the own-price channel for pork and legumes.

These results yield an interesting comparison to the literature: Shapiro (2020) finds a negative correlation between the emissions intensity of an industry and tariffs, meaning that greener products have higher tariffs. My findings in this paper contrast this result. I find larger tariffs for dirtier products in Home, implying that free trade would increase emissions in this setting. Bellora et al. (2025) find that fossil fuels are the main reason for the bias with their low tariffs, and adding agriculture to the sectors leads to a much smaller bias.

6.4 Reducing meat consumption

The next set of counterfactuals considers a reduction in meat consumption. As meat is a large contributor to emissions, reducing meat consumption might lead to high savings in emissions. Due to the nesting structure, removing a product from the choice set leads to a uniform increase in the quantity consumed of each other product. So each other food type is considered out-of-nest, leading to substitution with the same elasticity. Therefore, the quantities consumed increase in all scenarios, leading to a fairly consistent increase in emissions across food products.

6.4.1 No beef

The no beef scenario in the middle-left panel of Figure 6 considers a shutdown of beef consumption. Since beef has the highest emissions intensity of all meat types, a lot can be achieved by just refraining from eating beef. The total emissions decrease by around 14%. The emissions for each other food type increase by a few percentage points.

6.4.2 Vegetarianism

The vegetarian scenario[20] excludes all meat and fish and is displayed in the center panel of Figure 6. Each food type's emissions increases by around 8%. Because meat products have higher emissions than vegetarian products, the total effect is a decrease in emissions of around 20%.

6.4.3 No beef no cheese

Refraining from beef and cheese consumption in the middle-right panel of Figure 6 results in the highest decrease of overall emissions with around 35%. Each food type that remains in the choice set increases its emissions by 8%. Because beef and cheese are the largest contributors to emissions (beef by emissions

[19]The price decreases are small and hard to see in the graph.

[20]A vegetarian diet includes cheese consumption, for which holding cattle is necessary. Ideally, one would model the cattle demand for the level of cheese production in a vegetarian scenario and allow to consume the meat of the dairy cows. This is outside of the scope of this paper.

intensity and cheese by quantity consumed), switching to other meat types decreases emissions strongly. This scenario yields the largest decrease in emissions, although only two food types are excluded.

6.5 Pigouvian taxes

The following set of counterfactuals explores Pigouvian taxes, that is taxes that are proportional to the emissions intensity of a product. Pigouvian taxation a commonly proposed policy tool when externalities are present. In the context of this discrete choice model, all prices increase with Pigouvian taxes as all products have a certain level of emissions intensity[21]. I choose 50, 100, and 150 EUR/tCO_2e for the tax levels. The EU ETS price hit 100 EUR/tCO_2e in March 2023 and decreased again thereafter. Therefore, I provide a band around that value.

The lower three panels of Figure 6 show the Pigouvian tax scenarios for the three levels of taxation. The results are qualitatively similar for all three tax levels and get more pronounced with higher levels of the tax. The higher the tax, the larger the decreases in emissions for each product. Overall, the Pigouvian taxes lead to a 2%, 4%, and 6% decrease in total emissions for each of the tax levels.

Figure 9 shows that Pigouvian taxes lead to an increase in the price of all products. For most of them the own-price channel dominates so that the quantity consumed decreases. For tofu and legumes however, the cross-price channel dominates and the quantities consumed increase despite an increase in prices. This happens because the price increase for these low-emission products is small compared to large polluters like beef and cheese.

7 Robustness and external validity

This section presents robustness and external validity checks. For robustness, I show results that allow for a different definition of the outside good and results on barcode level data. To convince the reader of the relevance of consumption-side policies, I show a scenario with a decrease in emissions intensities without consumption changes. For external validity, I show how the results depend on the production emissions intensity of Home relatively to other countries and how consumption patterns in Home compare to other countries.

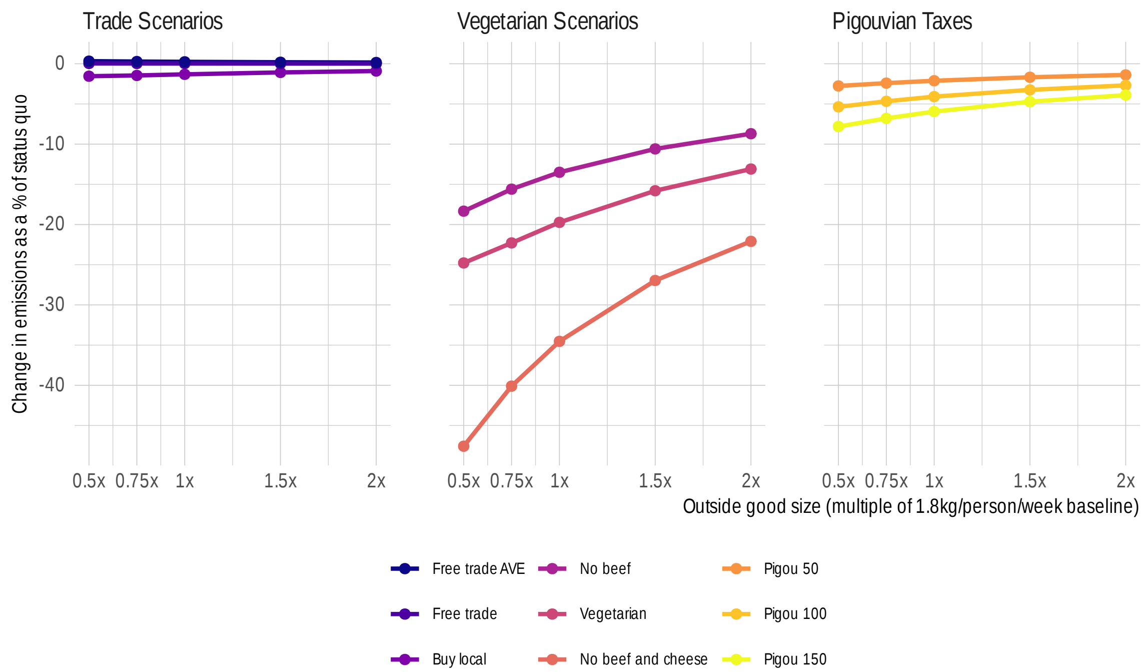
7.1 Outside good

As discussed in Section 2, I define the outside good as all other products consumers would purchase. Thereby, I make use of the empirical pattern that most humans eat around 1.8 kg of food every day to define the market size. In Figure 10, I show how the results would change for different levels of scaling of the outside good. The results are similar for the Trade scenarios and the Pigouvian Taxes. For the Vegetarian scenarios,

[21] In a general equilibrium model, the tax revenues would be rebated lump-sum to the households. This mechanism is not modeled in this setup.

having a smaller outside good share (scaled by 0.5 for example) leads to a larger decrease in emissions. The intuition is that the counterfactuals mainly change substitution within the inside goods, so if their share is larger, the effect on emissions is going to be larger. Qualitatively the results are similar.

Figure 10: Results with Outside Good Variation



Note: Each line is a counterfactual scenario and the y-axis is the change in total GHG emissions as a % of status-quo emissions. The x-axis rescales the size of the outside good relative to its baseline of 1.8 kg per person per day (e.g. 0.5x to 2x). Panels group the scenarios into trade, vegetarian, and Pigouvian tax policies.

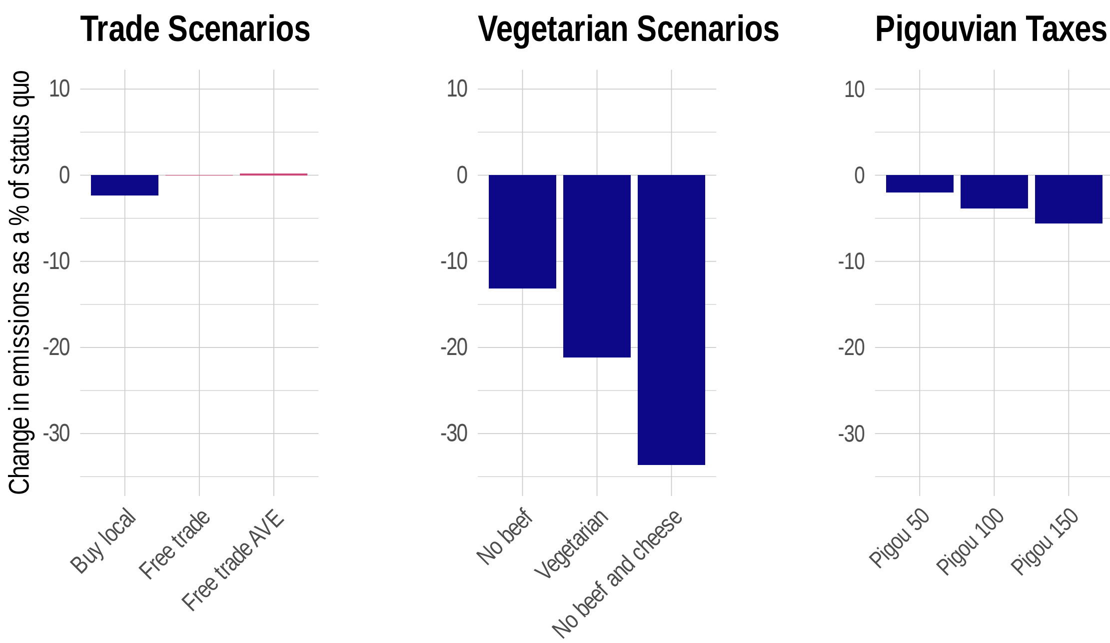
7.2 Barcode level results

Figure 11 shows the results for each counterfactual with barcode-level data, specifically with the regression from column 2 of Table 12. The results are similar in magnitude compared to the main specification.

7.3 Status quo with better production

Most of the trade and environment literature has focused on the supply side (Cherniwchan et al., 2017). In that spirit, the emissions of the agricultural sector could be reduced by improving production technologies in terms of their ecological efficiency. Gauging the effect of improved production would require a model of meat production and the corresponding emissions intensities. To give a sense of the magnitudes of such a policy, I calculate a naive improved production counterfactual. Keeping consumption shares of different food

Figure 11: Results with Barcode Level Data



Note: As in Figure 5, each bar is a counterfactual scenario and the y-axis is the change in total GHG emissions as a % of status-quo emissions, but here the demand system is estimated on barcode-level data (column 2 of Table 12).

Table 7: Status Quo with Best Production Counterfactual

	Quantity (in t)			Emissions (in tCO2e)		
Type	Before	After	Δ%	Before	After	Δ%
Beef	2,392	2,392	0	272,361	227,504	-16.47
Cheese	9,283	9,283	0	450,617	375,756	-16.61
Eggs	3,301	3,301	0	19,251	19,106	-0.75
Fish	1,079	1,079	0	40,692	21,497	-47.17
Lamb	364	364	0	35,319	28,654	-18.87
Legumes	662	662	0	1,898	1,560	-17.78
Pork	4,140	4,140	0	77,230	60,522	-21.63
Poultry	4,376	4,376	0	45,855	44,493	-2.97
Tofu	313	313	0	1,233	1,233	0
Veal	82	82	0	5,458	4,432	-18.8
Outside Good	137,123	137,123	0	893,054	893,054	0
Total	163,115	163,115	0	1,842,968	1,677,812	-8.96

Note.— This table shows a counterfactual in which I only change the production emissions to the best emissions for each food type. The values are for an average year per 1 mn consumers. The 'before' column refers to the status quo, the 'after' column to the counterfactual values. Rows are food types, 'Outside Good' is all other food and 'Total' is the sum across rows. Δ% is the percent change from the 'before' to the 'after' column.

types constant and assuming that production emissions are the lowest for each food type, I find that overall emissions decrease by around 9% in Table 7.

The assumption that all our food can be produced with the lowest emissions intensity for each food type is very strong. While it is possible to improve emissions intensities with better production practices, a part of emissions intensity of food production is related to land characteristics and cannot be adapted easily. These estimates on reducing agriculture's impact from the supply side represent an upper bound. Still, the question remains whether it would be possible to produce the same quantities with simply changing the production emissions. If the best possible emissions technology would imply lower yields, we might produce lower quantities on the same fields. Thus, we would need more land to produce the same amount of food which leads to higher emissions intensities. From a pricing perspective, it is unclear how ecologically efficient production maps into prices. Price changes depend on the competitiveness of the market. If the best production technology implies moving to a monopoly, prices might increase. Therefore, the true impact of improving production emissions intensities remains unclear.

Compared to this naive supply side counterfactual, all scenarios that imply reducing meat consumption yield a larger decrease in emissions. This implies that changing consumer behavior is an important factor for reducing the emissions of agriculture.

Table 8: Buy Local with Best Production Counterfactual

	Quantity (in t)			Emissions (in tCO2e)		
Type	Before	After	Δ%	Before	After	Δ%
Beef	2,392	2,316	-3.19	272,361	219,847	-19.28
Cheese	9,283	9,068	-2.32	450,617	366,588	-18.65
Eggs	3,301	3,087	-6.47	19,251	17,043	-11.47
Fish	1,079	925	-14.22	40,692	18,130	-55.45
Lamb	364	288	-21.03	35,319	20,218	-42.76
Legumes	662	607	-8.37	1,898	1,395	-26.48
Pork	4,140	4,162	0.53	77,230	60,840	-21.22
Poultry	4,376	4,367	-0.19	45,855	44,390	-3.2
Tofu	313	288	-7.9	1,233	1,098	-10.97
Veal	82	82	0.47	5,458	4,452	-18.43
Outside Good	137,123	137,925	0.58	893,054	898,273	0.58
Total	163,115	163,115	0	1,842,968	1,652,274	-10.35

Note.— This table shows how the buy local scenario would change if Home was the country with the best production emissions. The values are for an average year per 1 mn consumers. The 'before' column refers to the buy local scenario, the 'after' column to the counterfactual values. Rows are food types, 'Outside Good' is all other food and 'Total' is the sum across rows. Δ% is the percent change from the 'before' to the 'after' column.

7.4 Ecological efficiency

My results could depend on the production emissions intensity of Home relatively to other countries. I calculate the buy local scenario under the assumption that Home has the highest production emissions intensity and the lowest production emissions intensity for each food type. These scenarios are extreme and provide upper and lower bounds, respectively. There is no single country which has the highest or lowest production emissions intensity for each food type. The ecological production efficiency is diverse and depends not only on production practices, but also on natural endowments of countries.

Table 8 assumes that Home is the most efficient producer for each food type. Here the "Before" column shows the buy-local scenario with the modeled elasticities from Figure 7 and Appendix Table 16, not the status quo. Overall emissions would decrease by around 10%. This number is larger than the actual buy local scenario, but still smaller than all vegetarian scenarios. Table 9 shows that autarky would yield an increase in emissions of around 34% if Home was the ecologically least efficient producer in all food types. This suggests that buying local might lead to a small decrease in emissions for greener countries, and a large increase in emissions for dirtier countries. Thus, vegetarianism and Pigouvian taxes seem like more effective policies for decreasing greenhouse gas emissions from food consumption.

7.5 Consumption patterns across countries

The consumption patterns of Home could be different from other countries. Most of the literature attributes differences in consumption patterns to different prices and product characteristics. One exception is Dubois et al. (2014) who do find evidence for different preferences between consumers from the US, the UK, and

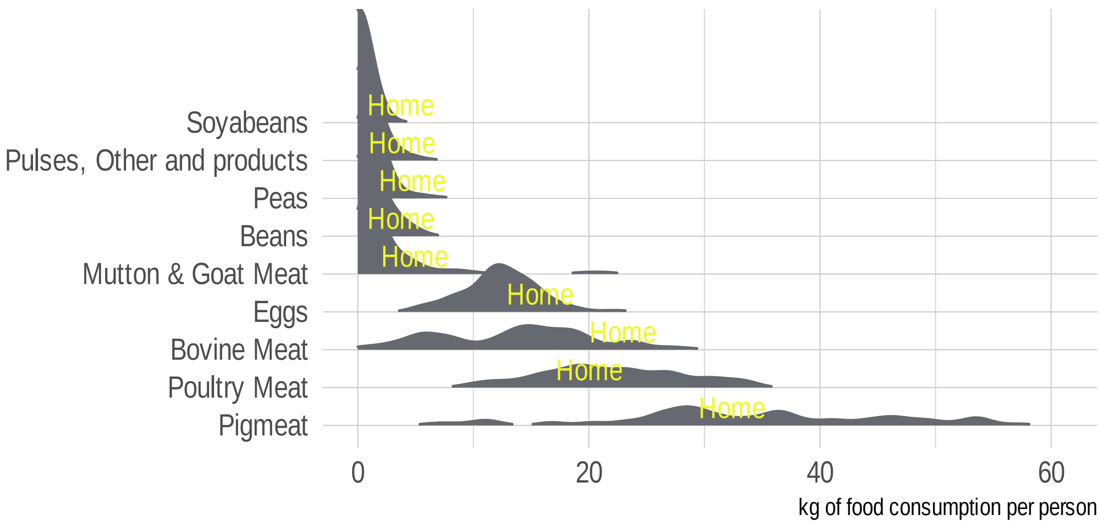
Table 9: Buy Local with Worst Production Counterfactual

	Quantity (in t)			Emissions (in tCO2e)		
Type	Before	After	Δ%	Before	After	Δ%
Beef	2,392	2,316	-3.19	272,361	787,261	189.05
Cheese	9,283	9,068	-2.32	450,617	444,794	-1.29
Eggs	3,301	3,087	-6.47	19,251	17,206	-10.62
Fish	1,079	925	-14.22	40,692	84,378	107.36
Lamb	364	288	-21.03	35,319	29,598	-16.2
Legumes	662	607	-8.37	1,898	1,880	-0.95
Pork	4,140	4,162	0.53	77,230	80,249	3.91
Poultry	4,376	4,367	-0.19	45,855	113,082	146.61
Tofu	313	288	-7.9	1,233	1,098	-10.97
Veal	82	82	0.47	5,458	8,870	62.53
Outside Good	137,123	137,925	0.58	893,054	898,273	0.58
Total	163,115	163,115	0	1,842,968	2,466,689	33.84

Note.— This table shows how the buy local scenario would change if Home was the country with the worst production emissions. The values are for an average year per 1 mn consumers. The 'before' column refers to the buy local scenario, the 'after' column to the counterfactual values. Rows are food types, 'Outside Good' is all other food and 'Total' is the sum across rows. Δ% is the percent change from the 'before' to the 'after' column.

France in terms of caloric intake. Still, their estimates for the preferences of meat and dairy in those countries are similar. To gauge how representative Home is for Europe, I compare the average meat consumption patterns of different European countries using the quantity of domestic food supply data from the Food and Agricultural Organization. Figure 12 shows that Home is close to the mean for all available food types. Thus, Home is not an outlier compared to a large number of European countries.

Figure 12: Food Consumption across Europe



Note: Each row is a food type and each point is a country's annual per-capita consumption of that food over 2016–2019, in kg per person. The point for Home is labeled in every row.

8 Conclusion

This paper asks how much demand-side policies can reduce greenhouse gas emissions from food consumption. Combining barcode-level data from a European retailer with product-level emissions intensities (Poore and Nemecek, 2018), I estimate a nested logit demand system for protein-rich foods and embed it in a partial equilibrium model with a fixed-margin supply side. This structure lets me trace how policies that change costs pass through into prices, and how policies that change the choice set reallocate demand across foods and origins, while holding the retailer's margins fixed.

The central lesson is that consumer substitution determines whether a food policy helps or hurts the environment. Buying local, the measure European consumers most associate with sustainability, is the clearest example: a naive accounting exercise that holds quantities fixed and reassigns all production Home implies that emissions rise, because several imported varieties are cleaner than their domestic counterparts. Once the estimated elasticities are allowed to guide substitution away from imports, emissions fall, but only modestly. Free trade moves emissions in the opposite direction from the standard intuition: because beef, veal, and lamb carry the highest tariffs in Home, liberalization raises consumption of the most polluting foods and slightly increases emissions, whether tariffs or their ad-valorem-equivalent counterparts under binding quotas are used.

Policies that reshape the diet deliver the largest reductions. Vegetarianism lowers emissions by around 20%, while removing beef alone lowers them by about 14%. Because cheese is the second-largest source of emissions after beef, a scenario that removes both beef and cheese reduces emissions by roughly 35%—more than vegetarianism itself. Pigouvian carbon taxes proportional to each product's emissions intensity deliver more moderate reductions of about 2%, 4%, and 6% at 50, 100, and 150 EUR/tCO_2e, while leaving the full choice set available. All of these consumption- and tax-based policies outperform buying local. The answer to the question in the title is therefore "not to beef": cutting the most emissions-intensive foods, above all beef and cheese, is the most effective demand-side lever.

Beyond the policy rankings, estimating rather than calibrating substitution has two payoffs. First, the elasticities can reverse the sign of a counterfactual, as exemplified by the buy-local scenario: Accounting-based evaluations that ignore substitution can reach the wrong conclusion. Second, the ratio of across- to within-group substitution I estimate lies well below the 0.52–0.72 range used in quantitative trade and climate models (Correa-Dias et al., 2026; Costinot et al., 2016; Farrokhi and Pellegrina, 2023), implying a steeper substitution hierarchy and more resistance to cross-group substitution than those calibrations assume. These estimates can help discipline the demand side of such models.

Several exercises support the relevance and external validity of the results. Holding shares fixed and assigning each food its lowest observed emissions intensity implies a 9% reduction, a lower bound that leaves consumption-based policies with more room to cut. The buy-local result is sensitive to Home's relative cleanliness: emissions rise by 34% if Home is the dirtiest producer and fall by 10% if it is the cleanest, so local sourcing helps only for relatively green countries. Finally, Home's consumption patterns line up with those of other European countries once prices and product traits are accounted for, suggesting the findings extend to most developed European economies. Two robustness checks, namely varying the definition of the outside good and repeating the analysis at the barcode level, yield similar conclusions.

The analysis has limitations that point to avenues for future work. The model is partial equilibrium, so it abstracts from general equilibrium adjustments in production and from the lump-sum rebating of carbon-tax revenue, and the supply side is summarized by a fixed per-unit margin rather than an estimated pricing model due to data limitations. Extending the framework to a general equilibrium setting, or to a richer supply side in which producers and retailers respond to policy, would sharpen the aggregate implications. Even so, the results make a clear case that credible demand-side policy evaluation requires knowing how consumers substitute and that, on current evidence, moving away from beef and cheese does far more for the climate than buying local.

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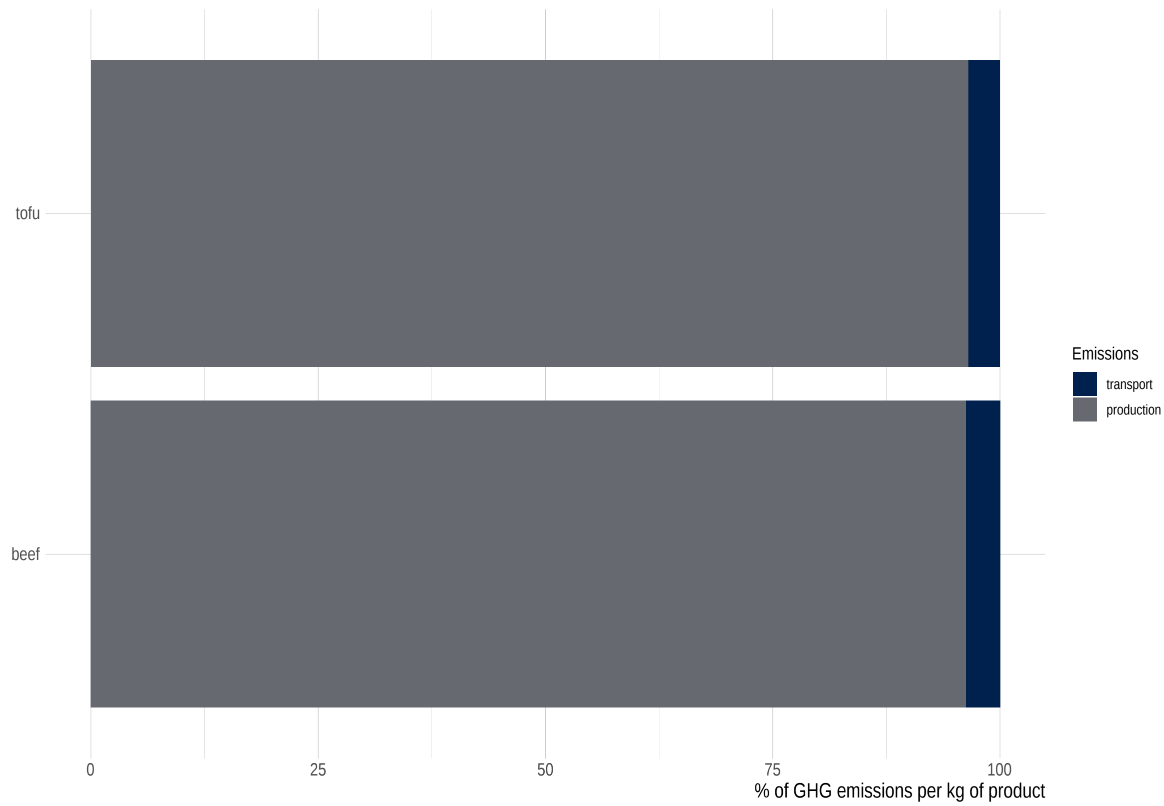
10 Appendix

10.1 Data treatment

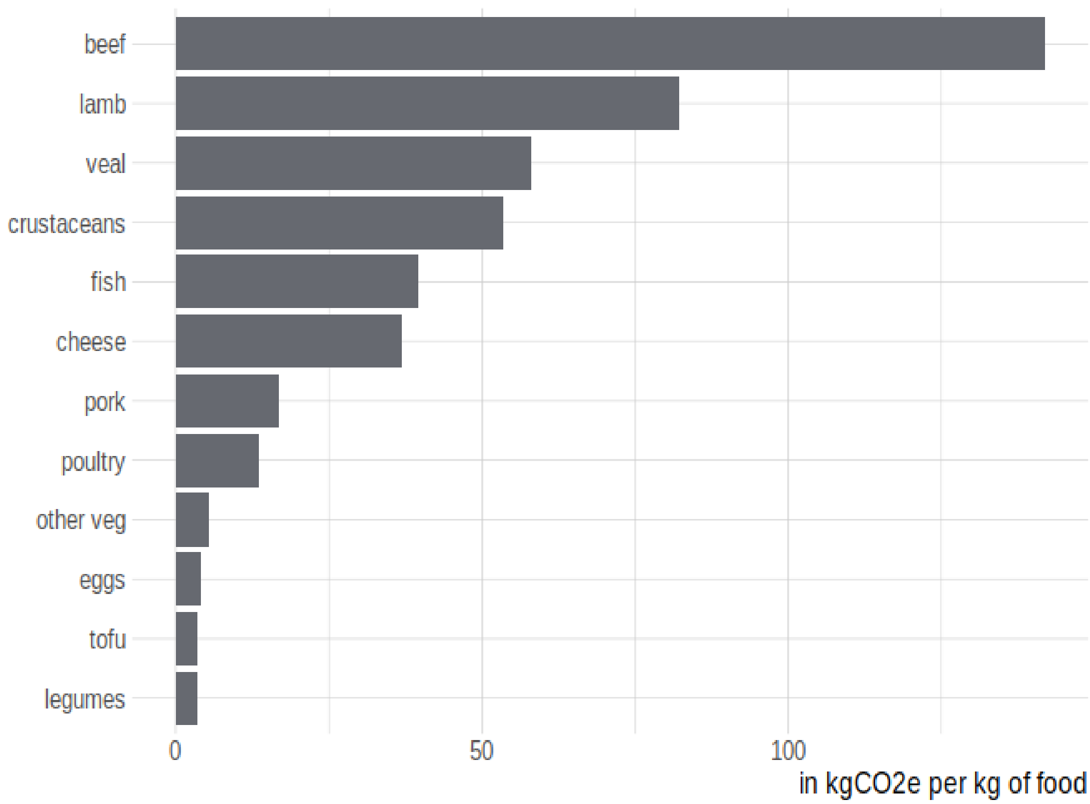
This section gives details on how to combine the production emissions from Poore and Nemecek (2018) with the supermarket data from the European retailer. Life cycle assessments (LCA) are a common method to analyze the environmental impact of a product. Typically, environmental scientists measure the emissions related to every part of a production process of a single product. As there exist many methodologies for LCAs, independent studies are typically not comparable. Poore and Nemecek (2018) combine data from 38,700 individual farms in a methodologically consistent manner and provide a worldwide overview of production emissions. I exclude the transport emissions from Poore and Nemecek, as this value combines transport emissions during the production process (e.g. transport of feed) with the transport to the final destination (e.g. importing beef from Argentina to Europe). I calculate the final destination transport emissions based on the actual country of origin for each product. The transport emissions during the production process are just a very small part of the overall emissions, so I omit them here. I do not observe whether a beef product stems from a dairy herd or a standard beef herd, so I use the standard beef herd emissions for all beef products. Meat from a dairy herd is rare, as most beef products originate from beef herds. Additionally, the consumption data contains veal products for which I do not observe emissions separately. Since calves are a by-product of milk production, I use the emissions from beef from the dairy herds for veal products. Finally, I include crustaceans in the fish category.

Figure 13: Total GHG emissions of imported beef and imported tofu



Note: Each horizontal bar shows the product with the median total emissions intensity among imported beef and among imported tofu, split into the share of per-kilogram GHG emissions from production and from transport to the retailer. The x-axis is the % of that product's total per-kg emissions.

Figure 14: Average GHG emissions of different food types excluding land use change



Note: This figure shows the average production GHG emissions intensity by food type, in kgCO_2e per kg of food, computed excluding emissions from land-use change.

10.2 Intuition example

This section gives a numerical example for the intuition of the own- and cross-price channels when several prices change. Here, we consider a simplified version of the model with a logit specification for two goods.

Table 10: Dummy Model Example

	p_j	p_l	s_j	s_l
1	10.0	10.0	0.00665	0.00665
2	8.0	10.0	0.01787	0.00657
3	8.0	3.0	0.01475	0.17973
4	9.9	3.0	0.00576	0.18138

Note: This table shows a numerical example for the own- and cross-price channels.

Table 10 illustrates several scenarios with α=-0.5. In row 1, products j and l have the same price and thus the same market shares. Decreasing the price of product j in row 2 yields an increase in the market share of product j, and a decrease in the market share of product l. This is the expected result form the own- and cross-price elasticities. In rows 3 and 4, I decrease the price of both goods compared to row 1. Product l experiences a much larger decrease than product j in both scenarios. In row 4, the market shares of both products increase through what I call the own-price channel. Note that s_j increases less than in row 2, because now also the price of product l decreased. This leads to a cross-price channel from the effect of the cross-price elasticity on the market share. In row 4, the price of product j only decreases a little, while the price of l decreases a lot. As a result, the market share of l increases, while the market share of good j *decreases*.

For both products, we have an own- and a cross-price channel that go in opposite directions. In row 4, product j had a small decrease in the price, and thus we would expect an increase in the market share from the own-price channel. However, product l had a large price decrease, from which we expect an increase in the market share of product l. The cross-price channel dominated over the own-price channel for product j, and thus its market share decreases.

10.3 Additional regression tables

Table 11: Nested Logit IV - All controls displayed

	Relative Market Share
	NL IV
Sales 1	2.133^{***} (0.066)
Sales 2	-1.999^{***} (0.074)
Sales 3	0.674^{***} (0.021)
European Retailer Brandname	3.152^{*} (1.654)
Chicken Brandname	4.069^{**} (1.653)
Sausage Brandname	6.811^{***} (1.660)
Other Brandname	2.634 (1.658)
No Brandname	2.896^{*} (1.655)
Cheese Brandname	2.384 (1.740)
Meat Substitute Brandname	1.776 (1.661)
Label Organic	-0.132^{***} (0.024)
Label Local	0.394^{***} (0.023)
Label Fish	-0.560^{***} (0.022)
Label Animal Welfare	0.164^{***} (0.046)
Label Allergy	8.948^{***} (0.693)
Label Grassfed	-0.060 (0.149)
Label Vegan	-0.828^{***} (0.077)
Label Vegetarian	0.238^{***} (0.079)
Price	-0.016^{***} (0.001)
Within Group MS	0.621^{***} (0.014)
Observations	39,262

Note: *p<0.1; **p<0.05; ***p<0.01

Includes quarter, year, region, continent and food type fixed effects, labels, and brands. This table shows a regression of the relative market share on prices, the within group market share, and other product characteristics.

Table 12: Nested Logit IV Barcode Level

	Relative Market Share			
	NL	NL IV	NL	NL IV
	(1)	(2)	(3)	(4)
Price	−0.001*** (0.00001)	−0.013*** (0.0002)	−0.001*** (0.00005)	−0.176*** (0.003)
Within Group MS	0.985*** (0.0001)	0.386*** (0.007)	0.977*** (0.0001)	0.403*** (0.020)
Sales 1	0.177*** (0.002)	2.589*** (0.028)		
Sales 2	−0.038*** (0.001)	−0.879*** (0.010)		
Sales 3	0.006*** (0.001)	−0.177*** (0.003)		
European Retailer Brand	0.015*** (0.002)	0.318*** (0.009)		
Chicken Brand	0.007*** (0.002)	0.177*** (0.010)		
Sausage Brand	0.013*** (0.002)	−0.204*** (0.010)		
Other Brand	0.020*** (0.002)	−0.271*** (0.009)		
No Brand	0.011*** (0.002)	−0.428*** (0.010)		
Cheese Brand	0.002 (0.002)	−0.052*** (0.009)		
Meat Substitute Brand	0.009*** (0.003)	0.128*** (0.012)		
Label Organic	−0.007*** (0.001)	0.134*** (0.003)		
Label Local	0.009*** (0.0004)	0.236*** (0.003)		
Label Fish	−0.037*** (0.001)	0.134*** (0.004)		
Label Animal Welfare	0.012*** (0.001)	0.153*** (0.005)		
Label Allergy	−0.010*** (0.002)	0.221*** (0.007)		
Label Grassfed	0.023*** (0.002)	−0.135*** (0.007)		
Label Vegan	−0.015*** (0.002)	−0.110*** (0.008)		
Label Vegetarian	−0.021*** (0.003)	0.038*** (0.010)		
Product FE	No	No	Yes	Yes
IV	None	Both	None	Both
Observations	2,597,846	2,597,846	2,597,846	2,597,846

Note: *p<0.1; **p<0.05; ***p<0.01

Includes quarter, year, region, continent and food type fixed effects, labels, and brands. This table shows a regression of the relative market share on prices, the within group market share, and other product characteristics.

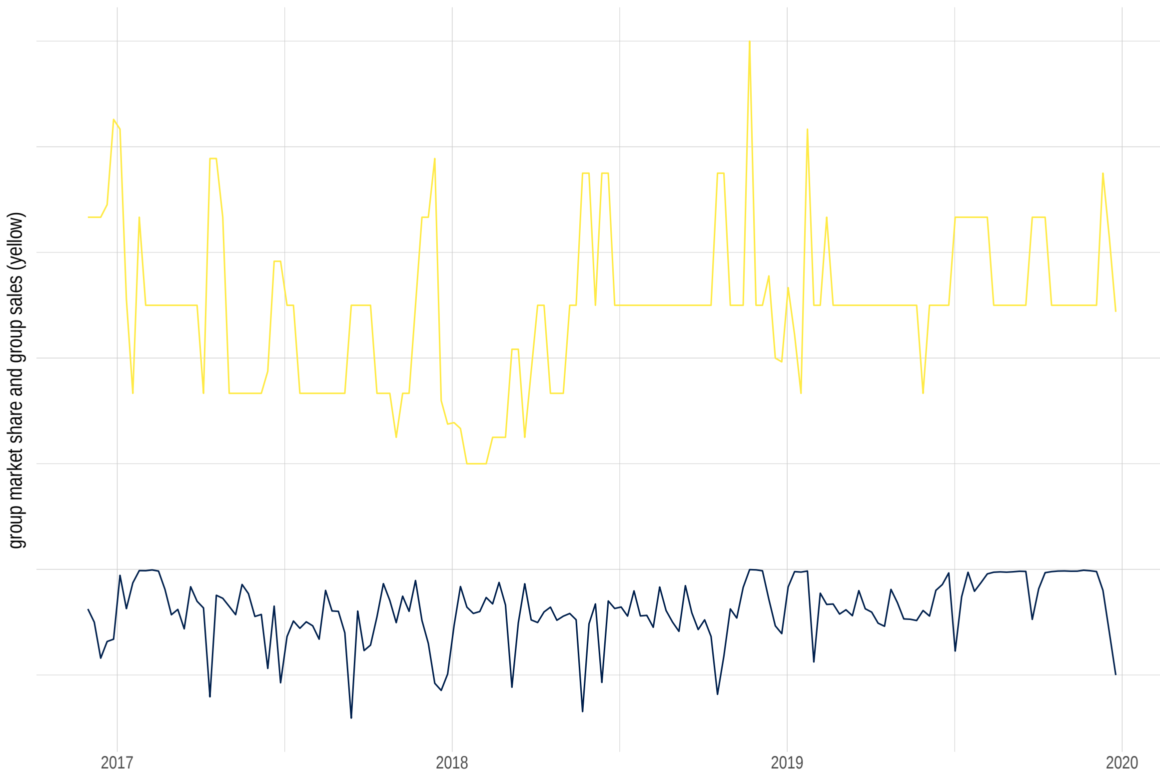
Table 13: Nested Logit Regression IV with Emissions

	Relative Market Share	
	Main	With Emissions
	(1)	(2)
Sales 1	2.133***	2.231***
	(0.066)	(0.066)
Sales 2	−1.999***	−1.840***
	(0.074)	(0.067)
Sales 3	0.674***	0.406***
	(0.021)	(0.013)
Label Organic	−0.132***	0.465***
	(0.024)	(0.031)
Label Local	0.394***	−0.033*
	(0.023)	(0.017)
Imported		−0.742***
		(0.028)
Emissions		−0.006***
		(0.0002)
Price	−0.016***	−0.015***
	(0.001)	(0.001)
Within Group MS	0.621***	0.692***
	(0.014)	(0.011)
Observations	39,262	39,262

Note: *p<0.1; **p<0.05; ***p<0.01

Includes quarter, year, region, and food type fixed effects, labels, and brands. This table shows a regression of the relative market share on prices, the within group market share, and other product characteristics. The first column includes continent fixed effects. The second column includes an imported dummy, and the greenhouse gas emissions related to each product.

Figure 15: Variation of within group market share



Note: Plots two series over the sample period (2017–2020): the within-group market share and the group-level sales proxy (in yellow) used to instrument it.

10.4 Elasticity formulas

The nested logit formula for calculating the own-price elasticity is given by:

$$\frac{\alpha p_{jrw}}{1-\sigma}\Big(1 - \sigma s_{j|g,rw} - (1-\sigma)s_{jrw}\Big),$$

where α is the price coefficient (negative), σ is the nesting parameter, p_{jrw} is the price of good j in market rw, s_{jrw} is the market share of good j, and $s_{j|g,rw}$ is the within-group market share of product j in nest g. The cross-price elasticity for products in the same nest is

$$-\frac{\alpha p_{krw}}{1-\sigma}\Big(\sigma s_{k|g,rw} + (1-\sigma)s_{krw}\Big),$$

	Type	Quantity		Emissions	
		in t	%	in tCO2e	in %
1	Beef	2392	1.47	272361	14.78
2	Cheese	9283	5.69	450617	24.45
3	Eggs	3301	2.02	19251	1.04
4	Fish	1079	0.66	40692	2.21
5	Lamb	364	0.22	35319	1.92
6	Legumes	662	0.41	1898	0.10
7	Pork	4140	2.54	77230	4.19
8	Poultry	4376	2.68	45855	2.49
9	Tofu	313	0.19	1233	0.07
10	Veal	82	0.05	5458	0.30
11	Outside Good	137123	84.07	893054	48.46
12	Total	163115	100.00	1842968	100.00

Note:
This table shows the consumption patterns in the status quo. The quantity is expressed in tons per 1 mn consumers.

Table 14: Status Quo

and the cross-price elasticity for products in different nests is the standard logit cross-price elasticity:

$$-\alpha p_{krw} s_{krw}.$$

10.5 Additional Tables

Table 15: Naive Buy Local Counterfactual

Type	Quantity (in t) Before	After	Δ%	Emissions (in tCO2e) Before	After	Δ%
Beef	2,392	2,392	0	272,361	271,458	-0.33
Cheese	9,283	9,283	0	450,617	455,374	1.06
Eggs	3,301	3,301	0	19,251	18,395	-4.44
Fish	1,079	1,079	0	40,692	41,348	1.61
Lamb	364	364	0	35,319	37,482	6.12
Legumes	662	662	0	1,898	1,881	-0.9
Pork	4,140	4,140	0	77,230	77,253	0.03
Poultry	4,376	4,376	0	45,855	44,473	-3.01
Tofu	313	313	0	1,233	1,192	-3.34
Veal	82	82	0	5,458	5,435	-0.42
Outside Good	137,123	137,123	0	893,054	893,054	0
Total	163,115	163,115	0	1,842,968	1,847,345	0.24

Note.— This table shows a naive buy local scenario, assuming that all imported food types will be produced in Home in an autarky scenario and ignoring the estimated cross-price elasticities. The values are for an average year per 1 mn consumers. The 'before' column refers to the status quo, the 'after' column to the counterfactual values. Rows are food types, 'Outside Good' is all other food and 'Total' is the sum across rows. Δ% is the percent change from the 'before' to the 'after' column.

Table 16: Buy Local Counterfactual

Type	Quantity (in t) Before	After	Δ%	Emissions (in tCO2e) Before	After	Δ%
Beef	2,392	2,316	-3.19	272,361	262,809	-3.51
Cheese	9,283	9,068	-2.32	450,617	444,794	-1.29
Eggs	3,301	3,087	-6.47	19,251	17,206	-10.62
Fish	1,079	925	-14.22	40,692	35,685	-12.3
Lamb	364	288	-21.03	35,319	29,598	-16.2
Legumes	662	607	-8.37	1,898	1,723	-9.2
Pork	4,140	4,162	0.53	77,230	77,660	0.56
Poultry	4,376	4,367	-0.19	45,855	44,390	-3.2
Tofu	313	288	-7.9	1,233	1,098	-10.97
Veal	82	82	0.47	5,458	5,461	0.05
Outside Good	137,123	137,925	0.58	893,054	898,273	0.58
Total	163,115	163,115	0	1,842,968	1,818,697	-1.32

Note.— This table shows a buy local counterfactual in which only domestic products can be purchased. The values are for an average year per 1 mn consumers. The 'before' column refers to the status quo, the 'after' column to the counterfactual values. Rows are food types, 'Outside Good' is all other food and 'Total' is the sum across rows. Δ% is the percent change from the 'before' to the 'after' column.

Table 17: Free Trade Tariff Counterfactual

	Quantity (in t)			Emissions (in tCO2e)		
Type	Before	After	Δ%	Before	After	Δ%
Beef	2,392	2,396	0.16	272,361	272,856	0.18
Cheese	9,283	9,283	-0.01	450,617	450,594	-0.01
Eggs	3,301	3,304	0.09	19,251	19,277	0.14
Fish	1,079	1,079	-0.01	40,692	40,690	-0.01
Lamb	364	365	0.14	35,319	35,361	0.12
Legumes	662	662	0.02	1,898	1,898	0.02
Pork	4,140	4,140	0	77,230	77,229	0
Poultry	4,376	4,376	0	45,855	45,867	0.03
Tofu	313	313	-0.01	1,233	1,233	-0.01
Veal	82	82	0.1	5,458	5,472	0.26
Outside Good	137,123	137,116	-0.01	893,054	893,009	-0.01
Total	163,115	163,115	0	1,842,968	1,843,485	0.03

Note.— This table shows a free trade counterfactual in which the current tariff rates are subtracted from the product price. The values are for an average year per 1 mn consumers. The 'before' column refers to the status quo, the 'after' column to the counterfactual values. Rows are food types, 'Outside Good' is all other food and 'Total' is the sum across rows. Δ% is the percent change from the 'before' to the 'after' column.

Table 18: Free Trade Ad Valorem Equivalent Counterfactual

	Quantity (in t)			Emissions (in tCO2e)		
Type	Before	After	Δ%	Before	After	Δ%
Beef	2,392	2,405	0.56	272,361	275,401	1.12
Cheese	9,283	9,281	-0.03	450,617	450,489	-0.03
Eggs	3,301	3,305	0.14	19,251	19,295	0.23
Fish	1,079	1,079	-0.03	40,692	40,678	-0.03
Lamb	364	379	4.08	35,319	36,515	3.38
Legumes	662	662	0	1,898	1,898	0
Pork	4,140	4,139	-0.02	77,230	77,211	-0.03
Poultry	4,376	4,385	0.22	45,855	46,387	1.16
Tofu	313	312	-0.03	1,233	1,233	-0.03
Veal	82	82	0.07	5,458	5,470	0.23
Outside Good	137,123	137,085	-0.03	893,054	892,802	-0.03
Total	163,115	163,115	0	1,842,968	1,847,380	0.24

Note.— This table shows a free trade counterfactual in which the ad valorem equivalent of tariff rate quotas is subtracted from the product price. The values are for an average year per 1 mn consumers. The 'before' column refers to the status quo, the 'after' column to the counterfactual values. Rows are food types, 'Outside Good' is all other food and 'Total' is the sum across rows. Δ% is the percent change from the 'before' to the 'after' column.

Table 19: No Beef Counterfactual

	Quantity (in t)			Emissions (in tCO2e)		
Type	Before	After	Δ%	Before	After	Δ%
Beef	2,392	0	-100	272,361	0	-100
Cheese	9,283	9,425	1.53	450,617	457,500	1.53
Eggs	3,301	3,351	1.52	19,251	19,544	1.52
Fish	1,079	1,095	1.5	40,692	41,303	1.5
Lamb	364	370	1.48	35,319	35,844	1.49
Legumes	662	672	1.52	1,898	1,927	1.52
Pork	4,140	4,204	1.53	77,230	78,411	1.53
Poultry	4,376	4,442	1.51	45,855	46,548	1.51
Tofu	313	317	1.51	1,233	1,252	1.51
Veal	82	83	1.56	5,458	5,543	1.56
Outside Good	137,123	139,156	1.48	893,054	906,292	1.48
Total	163,115	163,115	0	1,842,968	1,594,164	-13.5

Note.— This table shows a no beef counterfactual in which all other products can be purchased, except for beef. The values are for an average year per 1 mn consumers. The 'before' column refers to the status quo, the 'after' column to the counterfactual values. Rows are food types, 'Outside Good' is all other food and 'Total' is the sum across rows. Δ% is the percent change from the 'before' to the 'after' column.

Table 20: Vegetarian Counterfactual

	Quantity (in t)			Emissions (in tCO2e)		
Type	Before	After	Δ%	Before	After	Δ%
Beef	2,392	0	-100	272,361	0	-100
Cheese	9,283	10,062	8.39	450,617	488,416	8.39
Eggs	3,301	3,578	8.4	19,251	20,870	8.41
Fish	1,079	0	-100	40,692	0	-100
Lamb	364	0	-100	35,319	0	-100
Legumes	662	718	8.38	1,898	2,057	8.37
Pork	4,140	0	-100	77,230	0	-100
Poultry	4,376	0	-100	45,855	0	-100
Tofu	313	339	8.32	1,233	1,336	8.31
Veal	82	0	-100	5,458	0	-100
Outside Good	137,123	148,419	8.24	893,054	966,618	8.24
Total	163,115	163,115	0	1,842,968	1,479,296	-19.73

Note.— This table shows a vegetarian counterfactual in which no meat products can be purchased. The values are for an average year per 1 mn consumers. The 'before' column refers to the status quo, the 'after' column to the counterfactual values. Rows are food types, 'Outside Good' is all other food and 'Total' is the sum across rows. Δ% is the percent change from the 'before' to the 'after' column.

Table 21: No Beef No Cheese Counterfactual

	Quantity (in t)			Emissions (in tCO2e)		
Type	Before	After	Δ%	Before	After	Δ%
Beef	2,392	0	-100	272,361	0	-100
Cheese	9,283	0	-100	450,617	0	-100
Eggs	3,301	3,562	7.92	19,251	20,774	7.92
Fish	1,079	1,162	7.74	40,692	43,838	7.73
Lamb	364	393	7.93	35,319	38,124	7.94
Legumes	662	714	7.82	1,898	2,046	7.82
Pork	4,140	4,464	7.82	77,230	83,272	7.82
Poultry	4,376	4,717	7.8	45,855	49,429	7.79
Tofu	313	337	7.84	1,233	1,330	7.84
Veal	82	89	8.13	5,458	5,901	8.12
Outside Good	137,123	147,676	7.7	893,054	961,785	7.7
Total	163,115	163,115	0	1,842,968	1,206,499	-34.53

Note.— This table shows a no beef no cheese counterfactual in which all other products can be purchased, except for beef and cheese. The values are for an average year per 1 mn consumers. The 'before' column refers to the status quo, the 'after' column to the counterfactual values. Rows are food types, 'Outside Good' is all other food and 'Total' is the sum across rows. Δ% is the percent change from the 'before' to the 'after' column.

Table 22: 50EUR/tCO2 Pigouvian Tax Counterfactual

	Quantity (in t)			Emissions (in tCO2e)		
Type	Before	After	Δ%	Before	After	Δ%
Beef	2,392	2,196	-8.21	272,361	249,355	-8.45
Cheese	9,283	8,974	-3.34	450,617	435,521	-3.35
Eggs	3,301	3,301	0	19,251	19,248	-0.01
Fish	1,079	1,052	-2.5	40,692	39,562	-2.78
Lamb	364	339	-6.98	35,319	32,823	-7.07
Legumes	662	664	0.23	1,898	1,902	0.23
Pork	4,140	4,098	-1.01	77,230	76,450	-1.01
Poultry	4,376	4,359	-0.37	45,855	45,642	-0.46
Tofu	313	313	0.14	1,233	1,235	0.14
Veal	82	78	-4.68	5,458	5,201	-4.71
Outside Good	137,123	137,742	0.45	893,054	897,083	0.45
Total	163,115	163,115	0	1,842,968	1,804,021	-2.11

Note.— This table shows a Pigouvian tax counterfactual with a 50EUR/tCO2e tax on all products. The values are for an average year per 1 mn consumers. The 'before' column refers to the status quo, the 'after' column to the counterfactual values. Rows are food types, 'Outside Good' is all other food and 'Total' is the sum across rows. Δ% is the percent change from the 'before' to the 'after' column.

Table 23: 100EUR/tCO2 Pigouvian Tax Counterfactual

	Quantity (in t)			Emissions (in tCO2e)		
Type	Before	After	Δ%	Before	After	Δ%
Beef	2,392	2,015	-15.74	272,361	228,385	-16.15
Cheese	9,283	8,672	-6.58	450,617	420,845	-6.61
Eggs	3,301	3,300	-0.02	19,251	19,242	-0.05
Fish	1,079	1,025	-4.95	40,692	38,464	-5.47
Lamb	364	315	-13.49	35,319	30,498	-13.65
Legumes	662	665	0.44	1,898	1,906	0.44
Pork	4,140	4,056	-2.03	77,230	75,663	-2.03
Poultry	4,376	4,342	-0.76	45,855	45,422	-0.94
Tofu	313	313	0.27	1,233	1,236	0.26
Veal	82	74	-9.15	5,458	4,955	-9.21
Outside Good	137,123	138,335	0.88	893,054	900,948	0.88
Total	163,115	163,115	0	1,842,968	1,767,564	-4.09

Note.— This table shows a Pigouvian tax counterfactual with a 100EUR/tCO2e tax on all products. The values are for an average year per 1 mn consumers. The 'before' column refers to the status quo, the 'after' column to the counterfactual values. Rows are food types, 'Outside Good' is all other food and 'Total' is the sum across rows. Δ% is the percent change from the 'before' to the 'after' column.

Table 24: 150EUR/tCO2 Pigouvian Tax Counterfactual

	Quantity (in t)			Emissions (in tCO2e)		
Type	Before	After	Δ%	Before	After	Δ%
Beef	2,392	1,850	-22.66	272,361	209,246	-23.17
Cheese	9,283	8,380	-9.73	450,617	406,585	-9.77
Eggs	3,301	3,299	-0.06	19,251	19,232	-0.1
Fish	1,079	1,000	-7.35	40,692	37,398	-8.09
Lamb	364	293	-19.55	35,319	28,335	-19.78
Legumes	662	666	0.63	1,898	1,910	0.63
Pork	4,140	4,014	-3.06	77,230	74,869	-3.06
Poultry	4,376	4,324	-1.17	45,855	45,196	-1.44
Tofu	313	314	0.37	1,233	1,237	0.36
Veal	82	71	-13.44	5,458	4,720	-13.52
Outside Good	137,123	138,905	1.3	893,054	904,658	1.3
Total	163,115	163,115	0	1,842,968	1,733,386	-5.95

Note.— This table shows a Pigouvian tax counterfactual with a 150EUR/tCO2e tax on all products. The values are for an average year per 1 mn consumers. The 'before' column refers to the status quo, the 'after' column to the counterfactual values. Rows are food types, 'Outside Good' is all other food and 'Total' is the sum across rows. Δ% is the percent change from the 'before' to the 'after' column.